

Gravitation Theory *II*

Homework n°2.

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August 10, 2026

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1. Vielbein postulate

Problem 1.1. Show, in the coordinate basis, using the vielbein postulate $D_\mu e^a{}_\nu = 0$, that

$$R^a{}_b = \frac{1}{2} e^a{}_\rho E_b{}^\lambda R^\rho{}_{\lambda\mu\nu} dx^\mu \wedge dx^\nu, \quad (1)$$

$$T^a = \frac{1}{2} e^a{}_\rho T^\rho{}_{\mu\nu} dx^\mu \wedge dx^\nu, \quad (2)$$

with

$$R^\rho{}_{\lambda\mu\nu} = \partial_\mu \Gamma^\rho{}_{\nu\lambda} - \partial_\nu \Gamma^\rho{}_{\mu\lambda} + \Gamma^\rho{}_{\mu\sigma} \Gamma^\sigma{}_{\nu\lambda} - \Gamma^\rho{}_{\nu\sigma} \Gamma^\sigma{}_{\mu\lambda}, \quad (3)$$

$$T^\rho{}_{\mu\nu} = \Gamma^\rho{}_{\mu\nu} - \Gamma^\rho{}_{\nu\mu}. \quad (4)$$

Solution

To show this relations start from the Maurer-Cartán structure equations

$$R_b^a = d\omega_b^a + \omega_c^a \wedge \omega_b^c, \quad (5)$$

$$T^a = de^a + \omega_b^a \wedge e^b. \quad (6)$$

For the first expression, we can take (5) and write the spin connections in the coordinate basis. First we can see the exterior derivative term

$$d\omega_b^a = \partial_\mu \omega_{b\nu}^a dx^\mu \wedge dx^\nu,$$

and the connections just will be $\omega_b^a = \omega_{b\mu}^a dx^\mu$. So adding all, and making explicit the antisymmetry of the wedge product we have

$$\begin{aligned} R_b^a &= \frac{1}{2} \left(\partial_\mu \omega_{b\nu}^a - \partial_\nu \omega_{b\mu}^a \right) dx^\mu \wedge dx^\nu + \left(\omega_{c\mu}^a dx^\mu \right) \wedge \left(\omega_{b\nu}^c dx^\nu \right) \\ &= \frac{1}{2} \left(\partial_\mu \omega_{b\nu}^a - \partial_\nu \omega_{b\mu}^a \right) dx^\mu \wedge dx^\nu + \frac{1}{2} \left(\omega_{c\mu}^a \omega_{b\nu}^c - \omega_{c\nu}^a \omega_{b\mu}^c \right) dx^\mu \wedge dx^\nu \\ &= \frac{1}{2} \left[\partial_\mu \omega_{b\nu}^a - \partial_\nu \omega_{b\mu}^a + \omega_{c\mu}^a \omega_{b\nu}^c - \omega_{c\nu}^a \omega_{b\mu}^c \right] dx^\mu \wedge dx^\nu \\ &= \frac{1}{2} R_{b\mu\nu}^a dx^\mu \wedge dx^\nu. \end{aligned}$$

For other side, from the vielbein postulate

$$\partial_\mu e_\rho^a + \omega_{b\mu}^a e_\rho^b - \Gamma_{\mu\rho}^\lambda e_\lambda^a = 0, \quad (7)$$

we can apply a derivative on both sides of the equations

$$\partial_\nu \partial_\mu e_\rho^a = \partial_\nu \Gamma_{\mu\rho}^\lambda e_\lambda^a + \Gamma_{\mu\rho}^\lambda \partial_\nu e_\lambda^a - \partial_\nu \omega_{b\mu}^a e_\rho^b - \omega_{b\mu}^a \partial_\nu e_\rho^b, \quad (8)$$

and taking the same equation but with the indices $\mu \leftrightarrow \nu$

$$\partial_\mu \partial_\nu e_\rho^a = \partial_\mu \Gamma_{\nu\rho}^\lambda e_\lambda^a + \Gamma_{\nu\rho}^\lambda \partial_\mu e_\lambda^a - \partial_\mu \omega_{b\nu}^a e_\rho^b - \omega_{b\nu}^a \partial_\mu e_\rho^b \quad (9)$$

we can take (9) and rest to it (8)¹, so we will have

$$\begin{aligned} 0 &= \left(\partial_\mu \Gamma_{\nu\rho}^\lambda e_\lambda^a + \Gamma_{\nu\rho}^\lambda \partial_\mu e_\lambda^a - \partial_\mu \omega_{b\nu}^a e_\rho^b - \omega_{b\nu}^a \partial_\mu e_\rho^b \right) - \left(\partial_\nu \Gamma_{\mu\rho}^\lambda e_\lambda^a + \Gamma_{\mu\rho}^\lambda \partial_\nu e_\lambda^a - \partial_\nu \omega_{b\mu}^a e_\rho^b - \omega_{b\mu}^a \partial_\nu e_\rho^b \right) \\ &= \left(\partial_\mu \Gamma_{\nu\rho}^\lambda - \partial_\nu \Gamma_{\mu\rho}^\lambda \right) e_\lambda^a + \Gamma_{\nu\rho}^\lambda \partial_\mu e_\lambda^a - \Gamma_{\mu\rho}^\lambda \partial_\nu e_\lambda^a + \left(\partial_\mu \omega_{b\nu}^a - \partial_\nu \omega_{b\mu}^a \right) e_\rho^b + \omega_{b\mu}^a \partial_\nu e_\rho^b - \omega_{b\nu}^a \partial_\mu e_\rho^b. \end{aligned}$$

¹The left hand side is zero because the partial derivatives commute. This procedure is equivalent to study directly the commutator of covariant derivatives $[D_\mu, D_\nu]$ and identify from here expressions to replace with the vielbein postulate.

We can use again the vielbein postulate by replacing the partial derivatives with the connections terms (7). Term by term we will have

$$\Gamma_{\nu\rho}^\lambda \partial_\mu e_\lambda^a - \Gamma_{\mu\rho}^\lambda \partial_\nu e_\lambda^a = \Gamma_{\nu\rho}^\lambda \left(\Gamma_{\mu\lambda}^\sigma e_\sigma^a - \omega_{c\mu}^a e_\lambda^c \right) - \Gamma_{\mu\rho}^\lambda \left(\Gamma_{\nu\lambda}^\sigma e_\sigma^a - \omega_{c\nu}^a e_\lambda^c \right) \quad (10)$$

$$= \left(\Gamma_{\nu\rho}^\lambda \Gamma_{\mu\lambda}^\sigma - \Gamma_{\mu\rho}^\lambda \Gamma_{\nu\lambda}^\sigma \right) e_\sigma^a + \left(\Gamma_{\mu\rho}^\lambda \omega_{c\nu}^a - \Gamma_{\nu\rho}^\lambda \omega_{c\mu}^a \right) e_\lambda^c. \quad (11)$$

Doing the same use of the vielbein postulate with other two terms

$$\omega_{b\mu}^a \partial_\nu e_\rho^b - \omega_{b\nu}^a \partial_\mu e_\rho^b = \omega_{b\mu}^a \left(\Gamma_{\nu\rho}^\sigma e_\sigma^b - \omega_{c\nu}^b e_\rho^c \right) - \omega_{b\nu}^a \left(\Gamma_{\mu\rho}^\sigma e_\sigma^b - \omega_{c\mu}^b e_\rho^c \right) \quad (12)$$

$$= \left(\omega_{b\nu}^a \omega_{c\mu}^b - \omega_{b\mu}^a \omega_{c\nu}^b \right) e_\rho^c + \left(\omega_{b\mu}^a \Gamma_{\nu\rho}^\sigma - \omega_{b\nu}^a \Gamma_{\mu\rho}^\sigma \right) e_\sigma^b. \quad (13)$$

Replacing this two expressions in the equation (11) we have now

$$0 = \left(\partial_\mu \Gamma_{\nu\rho}^\lambda - \partial_\nu \Gamma_{\mu\rho}^\lambda \right) e_\lambda^a + \left(\Gamma_{\nu\rho}^\lambda \Gamma_{\mu\lambda}^\sigma - \Gamma_{\mu\rho}^\lambda \Gamma_{\nu\lambda}^\sigma \right) e_\sigma^a + \left(\Gamma_{\mu\rho}^\lambda \omega_{c\nu}^a - \Gamma_{\nu\rho}^\lambda \omega_{c\mu}^a \right) e_\lambda^c \quad (14)$$

$$+ \left(\partial_\mu \omega_{b\nu}^a - \partial_\nu \omega_{b\mu}^a \right) e_\rho^b + \left(\omega_{b\nu}^a \omega_{c\mu}^b - \omega_{b\mu}^a \omega_{c\nu}^b \right) e_\rho^c + \left(\omega_{b\mu}^a \Gamma_{\nu\rho}^\sigma - \omega_{b\nu}^a \Gamma_{\mu\rho}^\sigma \right) e_\sigma^b, \quad (15)$$

where we can reorganize the equation and rename dummy indices in order to obtain

$$\left(\partial_\mu \Gamma_{\nu\rho}^\sigma - \partial_\nu \Gamma_{\mu\rho}^\sigma + \Gamma_{\nu\rho}^\lambda \Gamma_{\mu\lambda}^\sigma - \Gamma_{\mu\rho}^\lambda \Gamma_{\nu\lambda}^\sigma \right) e_\sigma^a = \left(\partial_\mu \omega_{c\nu}^a - \partial_\nu \omega_{c\mu}^a + \omega_{b\mu}^a \omega_{c\nu}^b - \omega_{b\nu}^a \omega_{c\mu}^b \right) e_\rho^c. \quad (16)$$

From the right hand side of the equation we see exactly a expression which is proportional to that we have obtained from the Maurer-Cartán structure equation for the curvature (??), and in the left hand side the expression for the Riemann curvature tensor, so replacing first the left hand side

$$R_{\rho\mu\nu}^\sigma e_\sigma^a = \left(\partial_\mu \omega_{c\nu}^a - \partial_\nu \omega_{c\mu}^a + \omega_{b\mu}^a \omega_{c\nu}^b - \omega_{b\nu}^a \omega_{c\mu}^b \right) e_\rho^c / \cdot E_c^\rho \quad (17)$$

$$R_{\rho\mu\nu}^\sigma e_\sigma^a E_c^\rho = \left(\partial_\mu \omega_{c\nu}^a - \partial_\nu \omega_{c\mu}^a + \omega_{b\mu}^a \omega_{c\nu}^b - \omega_{b\nu}^a \omega_{c\mu}^b \right), \quad (18)$$

so replacing this in the expression

$$R_b^a = \frac{1}{2} \left[\partial_\mu \omega_{b\nu}^a - \partial_\nu \omega_{b\mu}^a + \omega_{c\mu}^a \omega_{b\nu}^c - \omega_{c\nu}^a \omega_{b\mu}^c \right] dx^\mu \wedge dx^\nu \quad (19)$$

we have that

$$R_b^a = \frac{1}{2} \left(R_{\rho\mu\nu}^\sigma e_\sigma^a E_c^\rho \right) dx^\mu \wedge dx^\nu, \quad (20)$$

which is the expression we wanted to obtain.

For the torsion expression we take again the Maurer-Cartán structure equation

$$T^a = de^a + \omega_b^a \wedge e^b, \quad (21)$$

and we write it in the coordinate basis

$$T^a = d(e_\nu^a dx^\nu) + (\omega_{b\mu}^a dx^\mu) \wedge (e_\nu^b dx^\nu) = \partial_\mu e_\nu^a dx^\mu \wedge dx^\nu + \omega_{b\mu}^a e_\nu^b dx^\mu \wedge dx^\nu \quad (22)$$

$$= \frac{1}{2} \left[\partial_\mu e_\nu^a - \partial_\nu e_\mu^a + \omega_{b\mu}^a e_\nu^b - \omega_{b\nu}^a e_\mu^b \right] dx^\mu \wedge dx^\nu, \quad (23)$$

where in the last line we just make explicit the antisymmetry of the wedge product. Here we directly replace the vielbein postulate (7) in the vielbein partial derivative terms

$$T^a = \frac{1}{2} \left[\left(-\omega_{b\mu}^a e_\nu^b + \Gamma_{\mu\nu}^\rho e_\rho^a + \omega_{b\nu}^a e_\mu^b - \Gamma_{\nu\mu}^\rho e_\rho^a \right) + \left(\omega_{b\mu}^a e_\nu^b - \omega_{b\nu}^a e_\mu^b \right) \right] dx^\mu \wedge dx^\nu \quad (24)$$

$$= \frac{1}{2} \left[(\omega_{b\nu}^a - \omega_{b\mu}^a) e_\mu^b + (\omega_{b\mu}^a - \omega_{b\nu}^a) e_\nu^b + (\Gamma_{\mu\nu}^\rho - \Gamma_{\nu\mu}^\rho) e_\rho^a \right] dx^\mu \wedge dx^\nu \quad (25)$$

$$= \frac{1}{2} \left[\Gamma_{\mu\nu}^\rho - \Gamma_{\nu\mu}^\rho \right] e_\rho^a dx^\mu \wedge dx^\nu, \quad (26)$$

Where we identify exactly the term of the torsion tensor, so we just replace it by the definition and obtain

$$T^a = \frac{1}{2} T_{\mu\nu}^\rho e_\rho^a dx^\mu \wedge dx^\nu. \quad (27)$$

2. Einstein-Hilbert action from Einstein-Cartán

Problem 2.1. *Recover the Einstein-Hilbert action, starting from Einstein-Cartan theory, by solving the algebraic equation for the connection*

$$\epsilon_{abcd} T^c \wedge e^d = 0. \quad (28)$$

Solution

The idea to solve this problem is to show how to obtain a metric dependent connection from a theory that establish the two concepts as an independent entities. This is the first step because the remaining calculation is to write the Einstein-Cartán (EC) action in the coordinated basis to make it look like the Einstein-Hilbert one.

As established previously, varying the Einstein-Cartán action with respect to ω gives the algebraic equation $\epsilon_{abcd} T^c \wedge e^d = 0$, whose unique solution is $T^a = 0$. We therefore take as our starting point the vanishing of torsion,

$$T^a = D e^a = d e^a + \hat{\omega}^a_b \wedge e^b = 0, \quad (29)$$

and solve it *explicitly* for the metric ($\hat{\omega}^{ab} = -\hat{\omega}^{ba}$) connection one-form, in the frame basis. We then pass, through the vielbein postulate, to the coordinate description and identify $\hat{\omega}(e)$ with the Levi-Civita (Christoffel) connection of $g_{\mu\nu} = \eta_{ab} e^a_\mu e^b_\nu$.

Frame indices $a, b, \dots$ are moved with $\eta_{ab} = \text{diag}(-1, +1, +1, +1)$ and coordinate indices with $g_{\mu\nu}$.

Because $\{e^a\}$ is a co-frame, the two-form $d e^a$ can be expanded in the basis $\{e^b \wedge e^c\}_{b < c}$. Following the proposed strategy, we *define* the anholonomy (structure) coefficients Ω^a_{bc} by

$$d e^a = \frac{1}{2} \Omega^a_{bc} e^b \wedge e^c, \quad \Omega^a_{bc} = -\Omega^a_{cb}, \quad (30)$$

where the antisymmetry in bc is a convention with no loss of generality, since only the $[bc]$ -part of any coefficient survives contraction with the antisymmetric $e^b \wedge e^c$.

To get the explicit form, compute $d e^a$ in the coordinated basis and then transform it into the orthonormal frame

$$\begin{aligned} d e^a &= d(e^a_\mu dx^\mu) = \partial_\nu e^a_\mu dx^\nu \wedge dx^\mu = \partial_\mu e^a_\nu dx^\mu \wedge dx^\nu \\ &= \partial_\mu e^a_\nu E_b^\mu E_c^\nu e^b \wedge e^c = \frac{1}{2} (E_b^\mu E_c^\nu - E_b^\nu E_c^\mu) \partial_\mu e^a_\nu e^b \wedge e^c, \end{aligned} \quad (31)$$

where in the last step the coefficient was antisymmetrized in $b \leftrightarrow c$. Comparing (31) with (30) and relabeling $\mu \leftrightarrow \nu$ in the second piece,

$$\boxed{\Omega_{bc}^a = (E_b^\mu E_c^\nu - E_b^\nu E_c^\mu) \partial_\mu e^a_\nu = E_b^\mu E_c^\nu (\partial_\mu e^a_\nu - \partial_\nu e^a_\mu)} \quad (32)$$

which is manifestly antisymmetric in bc , as required².

Now we insert (30) into the torsion-free condition (29):

$$0 = T^a = \frac{1}{2} \Omega_{bc}^a e^b \wedge e^c + \dot{\omega}_{b|c}^a e^c \wedge e^b, \quad (33)$$

where the index of the basis and the Lorentz form are distinguished by the notation with the "|" in $\omega_{b|c}^a$. In the second term we just relabel the pair $b \leftrightarrow c$, giving $\dot{\omega}_{c|b}^a e^b \wedge e^c$, whose antisymmetric part in bc is $\frac{1}{2}(\dot{\omega}_{c|b}^a - \dot{\omega}_{b|c}^a)$:

$$0 = T^a = \frac{1}{2} [\Omega_{bc}^a + \dot{\omega}_{c|b}^a - \dot{\omega}_{b|c}^a] e^b \wedge e^c. \quad (34)$$

The two-forms $\{e^b \wedge e^c\}_{b < c}$ are pointwise linearly independent, and the bracket is already antisymmetric in bc ; hence (34) is equivalent to the vanishing of the bracket,

$$\dot{\omega}_{b|c}^a - \dot{\omega}_{c|b}^a = \Omega_{bc}^a. \quad (35)$$

Lowering the index a with the Minkowski metric yields the relation

$$\boxed{\Omega_{abc} = \dot{\omega}_{ab|c} - \dot{\omega}_{ac|b}} \quad (36)$$

together with the two structural symmetries

$$\Omega_{abc} = -\Omega_{acb} \quad (\text{by (32)}), \quad \dot{\omega}_{ab|c} = -\dot{\omega}_{ba|c} \quad (\text{metricity}). \quad (37)$$

Equation (36) is a linear algebraic system for the $4 \times 6 = 24$ unknowns $\dot{\omega}_{ab|c}$ (antisymmetric in ab) in terms of the 24 known Ω_{abc} (antisymmetric in bc).

Now we isolate $\dot{\omega}_{ab|c}$ by the standard cyclic permutation of indices and adding all the equations in a convenient way. Write (36) three times, cyclically permuting the free triple $(a, b, c) \rightarrow (b, c, a) \rightarrow (c, a, b)$,

$$\Omega_{abc} = \dot{\omega}_{ab|c} - \dot{\omega}_{ac|b}, \quad (M1)$$

$$\Omega_{bca} = \dot{\omega}_{bc|a} - \dot{\omega}_{ba|c}, \quad (M2)$$

$$\Omega_{cab} = \dot{\omega}_{ca|b} - \dot{\omega}_{cb|a}. \quad (M3)$$

Form the combination (M1) + (M2) - (M3) and simplify the right-hand side using only the metricity antisymmetry, specifically $-\dot{\omega}_{ba|c} = +\dot{\omega}_{ab|c}$, $-\dot{\omega}_{ca|b} = +\dot{\omega}_{ac|b}$, $+\dot{\omega}_{cb|a} = -\dot{\omega}_{bc|a}$:

$$\begin{aligned} \Omega_{abc} + \Omega_{bca} - \Omega_{cab} &= (\dot{\omega}_{ab|c} - \dot{\omega}_{ac|b}) + (\dot{\omega}_{bc|a} - \dot{\omega}_{ba|c}) - (\dot{\omega}_{ca|b} - \dot{\omega}_{cb|a}) \\ &= \underbrace{\dot{\omega}_{ab|c} - \dot{\omega}_{ac|b}}_{=+\dot{\omega}_{ab|c}} + \underbrace{\dot{\omega}_{bc|a} - \dot{\omega}_{ba|c}}_{=+\dot{\omega}_{ab|c}} - \underbrace{\dot{\omega}_{ca|b} - \dot{\omega}_{cb|a}}_{=-\dot{\omega}_{bc|a}} \\ &= 2\dot{\omega}_{ab|c} + (-\dot{\omega}_{ac|b} + \dot{\omega}_{ac|b}) + (\dot{\omega}_{bc|a} - \dot{\omega}_{bc|a}) = 2\dot{\omega}_{ab|c}. \end{aligned} \quad (38)$$

Hence

$$\boxed{\dot{\omega}_{ab|c}(e) = \frac{1}{2}(\Omega_{abc} + \Omega_{bca} - \Omega_{cab})} \quad (39)$$

and the connection one-form is

$$\dot{\omega}_{ab} = \dot{\omega}_{ab|c} e^c = \frac{1}{2}(\Omega_{abc} + \Omega_{bca} - \Omega_{cab}) e^c. \quad (40)$$

²Equivalently, Ω_{bc}^a are minus the structure functions of the frame vector fields: $[e_b, e_c] = -\Omega_{bc}^a e_a$.

Consistency: it is a Lorentz connection. We must verify that (39) indeed satisfies $\dot{\omega}_{ab|c} = -\dot{\omega}_{ba|c}$, so that $\dot{\omega}^{ab} \in \mathfrak{so}(1,3)$. Exchanging $a \leftrightarrow b$ in (39) and using $\Omega_{bac} = -\Omega_{bca}$, $\Omega_{acb} = -\Omega_{abc}$, $\Omega_{cba} = -\Omega_{cab}$,

$$\dot{\omega}_{ba|c} = \frac{1}{2} \left(\Omega_{bac} + \Omega_{acb} - \Omega_{cba} \right) = \frac{1}{2} \left(-\Omega_{bca} - \Omega_{abc} + \Omega_{cab} \right) = -\dot{\omega}_{ab|c}. \quad (41)$$

By construction (39) also solves the master relation (36) (this is what (38) proves), and the solution is unique.

We now translate the frame result into the coordinate (holonomic) language. Introduce the affine connection $\Gamma^\lambda_{\mu\nu}$ through the *vielbein postulate*:

$$D_\mu e^a{}_\nu := \partial_\mu e^a{}_\nu + \dot{\omega}^a{}_{b\mu} e^b{}_\nu - \Gamma^\lambda_{\mu\nu} e^a{}_\lambda = 0. \quad (42)$$

Contracting (42) with $e_a{}^\rho$ and using $E_a{}^\rho e^a{}_\lambda = \delta^\rho_\lambda$ solves it algebraically for Γ ,

$$\boxed{\Gamma^\lambda_{\mu\nu} = e_a{}^\lambda \left(\partial_\mu e^a{}_\nu + \dot{\omega}^a{}_{b\mu} e^b{}_\nu \right)} \quad (43)$$

where $\dot{\omega}^a{}_{b\mu} = \dot{\omega}^a{}_{b|c} e^c{}_\mu$ is fixed by (39). It remains to show that this Γ is the Levi-Civita connection, i.e. that it is (i) symmetric and (ii) metric-compatible.

(i) Symmetry: We just antisymmetrize (43) in $\mu\nu$. Writing out the mixed-index torsion components,

$$T^a{}_{\mu\nu} = 2 \left(\partial_{[\mu} e^a{}_{\nu]} + \dot{\omega}^a{}_{b[\mu} e^{b]}{}_{\nu]} \right) = 2 \Gamma^\lambda_{[\mu\nu]} e^a{}_\lambda, \quad (44)$$

where the first equality is just the coordinate expansion of $T^a = de^a + \dot{\omega}^a{}_b \wedge e^b$ and the second uses (42). Since $T^a = 0$ (our starting point) and $e^a{}_\lambda$ is invertible,

$$\Gamma^\lambda_{[\mu\nu]} = 0 \quad \Longleftrightarrow \quad \Gamma^\lambda_{\mu\nu} = \Gamma^\lambda_{\nu\mu}. \quad (45)$$

(ii) Metric compatibility. Considering that $g_{\nu\rho} = \eta_{ab} e^a{}_\nu e^b{}_\rho$, we can eliminate $\partial_\mu e^a{}_\nu$ using (42), $\partial_\mu e^a{}_\nu = \Gamma^\lambda_{\mu\nu} e^a{}_\lambda - \dot{\omega}^a{}_{c\mu} e^c{}_\nu$. Starting from the covariant derivative of a symmetric rank 2 tensor

$$\begin{aligned} \nabla_\mu g_{\nu\rho} &= \partial_\mu g_{\nu\rho} - \Gamma^\lambda_{\mu\nu} g_{\lambda\rho} - \Gamma^\lambda_{\mu\rho} g_{\nu\lambda} \\ &= \eta_{ab} \left[(\Gamma^\lambda_{\mu\nu} e^a{}_\lambda - \dot{\omega}^a{}_{c\mu} e^c{}_\nu) e^b{}_\rho + e^a{}_\nu (\Gamma^\lambda_{\mu\rho} e^b{}_\lambda - \dot{\omega}^b{}_{c\mu} e^c{}_\rho) \right] - \Gamma^\lambda_{\mu\nu} g_{\lambda\rho} - \Gamma^\lambda_{\mu\rho} g_{\nu\lambda} \\ &= -(\dot{\omega}^a{}_{b\mu} e^b{}_\nu e^a{}_\rho + \dot{\omega}^a{}_{c\mu} e^a{}_\nu e^c{}_\rho) = -(\dot{\omega}_{cb\mu} + \dot{\omega}_{bc\mu}) e^b{}_\nu e^c{}_\rho = 0, \end{aligned} \quad (46)$$

where the Γ -terms cancel the last two terms identically (using $\eta_{ab} e^a{}_\lambda e^b{}_\rho = g_{\lambda\rho}$) and the final vanishing is precisely the metricity antisymmetry $\dot{\omega}_{cb\mu} = -\dot{\omega}_{bc\mu}$ of (37).

Until now, we have shown that the affine connection defined by the vielbein postulate, (43), is symmetric and metric-compatible,

$$\Gamma^\lambda_{\mu\nu} = \Gamma^\lambda_{\nu\mu}, \quad \nabla_\mu g_{\nu\rho} = 0, \quad g_{\mu\nu} = \eta_{ab} e^a{}_\mu e^b{}_\nu. \quad (47)$$

We now make explicit how these two properties *alone* fix Γ and yield the closed Christoffel expression. Expanding the covariant derivative of the metric,

$$\nabla_\mu g_{\nu\rho} = \partial_\mu g_{\nu\rho} - \Gamma^\sigma_{\mu\nu} g_{\sigma\rho} - \Gamma^\sigma_{\mu\rho} g_{\nu\sigma} = 0. \quad (48)$$

Now we introduce the fully lowered symbol

$$\Gamma_{\rho\mu\nu} := g_{\rho\sigma} \Gamma^\sigma_{\mu\nu}, \quad \Gamma_{\rho\mu\nu} = \Gamma_{\rho\nu\mu}, \quad (49)$$

which inherits the symmetry in its last two indices from $\Gamma^\sigma_{\mu\nu} = \Gamma^\sigma_{\nu\mu}$. Then (48) reads, after moving the derivative to one side,

$$\partial_\mu g_{\nu\rho} = \Gamma_{\rho\mu\nu} + \Gamma_{\nu\mu\rho}, \quad (50)$$

where we used $\Gamma^\sigma_{\mu\nu} g_{\sigma\rho} = \Gamma_{\rho\mu\nu}$ and $\Gamma^\sigma_{\mu\rho} g_{\nu\sigma} = \Gamma_{\nu\mu\rho}$. Equation (50) is the exact analogue of the master relation $\Omega_{abc} = \dot{\omega}_{ab|c} - \dot{\omega}_{ac|b}$: the left-hand side is now ∂g (the metric is not constant in coordinates), and the relative sign is + rather than - because $\Gamma_{\rho\mu\nu}$ is symmetric in the pair to be permuted, whereas $\dot{\omega}_{ab|c}$ was antisymmetric.

Now just write (50) the cyclic permutation of the indices $(\mu, \nu, \rho) \rightarrow (\nu, \rho, \mu) \rightarrow (\rho, \mu, \nu)$:

$$\partial_\mu g_{\nu\rho} = \Gamma_{\rho\mu\nu} + \Gamma_{\nu\mu\rho}, \quad (\text{A}) \quad (51)$$

$$\partial_\nu g_{\rho\mu} = \Gamma_{\mu\nu\rho} + \Gamma_{\rho\nu\mu}, \quad (\text{B}) \quad (52)$$

$$\partial_\rho g_{\mu\nu} = \Gamma_{\nu\rho\mu} + \Gamma_{\mu\rho\nu}. \quad (\text{C}) \quad (53)$$

Form (A)+(B)-(C) and simplify the right-hand side using only the symmetry $\Gamma_{\alpha\beta\gamma} = \Gamma_{\alpha\gamma\beta}$:

$$(\Gamma_{\rho\mu\nu} + \Gamma_{\rho\nu\mu}) + (\Gamma_{\nu\mu\rho} - \Gamma_{\nu\rho\mu}) + (\Gamma_{\mu\nu\rho} - \Gamma_{\mu\rho\nu}) = 2\Gamma_{\rho\mu\nu}, \quad (54)$$

using the symmetries of the indices ($\Gamma_{\rho\mu\nu} = \Gamma_{\rho\nu\mu}$), ($\Gamma_{\nu\mu\rho} = \Gamma_{\nu\rho\mu}$ and $\Gamma_{\mu\nu\rho} = \Gamma_{\mu\rho\nu}$). Hence

$$\partial_\mu g_{\nu\rho} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu} = 2\Gamma_{\rho\mu\nu}. \quad (55)$$

Solving (55) and contracting with $g^{\lambda\rho}$, using $\Gamma^\lambda_{\mu\nu} = g^{\lambda\rho} \Gamma_{\rho\mu\nu}$ and $g_{\nu\rho} = g_{\rho\nu}$,

$$\boxed{\Gamma^\lambda_{\mu\nu} = \frac{1}{2} g^{\lambda\rho} (\partial_\mu g_{\rho\nu} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu})} \quad (56)$$

which is the standard Christoffel expression. We never compare the frame formula $\dot{\omega}_{ab|c} = \frac{1}{2}(\Omega_{abc} + \Omega_{bca} - \Omega_{cab})$ against ∂g term by term. The argument is by uniqueness: the frame connection is antisymmetric ($\mathfrak{so}(1,3)$) and torsion-free by construction; through the vielbein postulate (43) these translate into metric compatibility and symmetry of Γ , as recorded in (47); and the fundamental theorem of (pseudo-)Riemannian geometry guarantees a *unique* affine connection with those two properties [1](Theorem 3.1.1).

Recovering the Einstein–Hilbert action

With the connection solved, $\omega = \dot{\omega}(e)$, the final step is purely algebraic: substitute the solution into the action and contract indices.

In general one may not feed the solution of some field equations back into the action and expect the correct dynamics for the remaining fields. It is allowed precisely when the eliminated field is *auxiliary*: its own equation of motion is algebraic in that field (no derivatives of it appear) and has a unique solution in terms of the others. This is exactly our situation, the equation for ω turned out to be $\epsilon_{abcd} T^c \wedge e^d = 0 \Leftrightarrow T^a = de^a + \omega^a_b \wedge e^b = 0$: it contains ω but not $d\omega$, so it is algebraic in ω and we solved it uniquely, $\omega = \dot{\omega}(e)$. The elimination lemma then states that, writing $\tilde{S}[e] := S[e, \dot{\omega}(e)]$,

$$\frac{\delta \tilde{S}}{\delta e} = \frac{\delta S}{\delta e} \Big|_{\dot{\omega}(e)} + \underbrace{\frac{\delta S}{\delta \omega} \Big|_{\dot{\omega}(e)}}_{=0} \frac{\delta \dot{\omega}}{\delta e} = \frac{\delta S}{\delta e} \Big|_{\dot{\omega}(e)}, \quad (57)$$

the chain-rule term vanishing because its first factor is zero by construction. Hence the stationary points of $\tilde{S}[e]$ are exactly those of the full problem $S[e, \omega]$ with $\omega = \dot{\omega}(e)$: the two variational problems are equivalent, and we may substitute freely.

Recall the Einstein–Cartan action (vacuum, $\Lambda = 0$),

$$S_{\text{EC}}[e, \omega] = \frac{1}{4\kappa} \int_M \epsilon_{abcd} e^a \wedge e^b \wedge R^{cd}[\omega]. \quad (58)$$

We need two identities. First, the volume form together with the four-form identity,

$$\text{vol} := e^0 \wedge e^1 \wedge e^2 \wedge e^3 = \sqrt{-g} d^4x, \quad e^a \wedge e^b \wedge e^c \wedge e^d = -\epsilon^{abcd} \text{vol}, \quad (59)$$

where the sign follows because both sides are totally antisymmetric in the four indices. Second, the contraction of two Levi-Civita symbols with the Minkowski determinant sign convention,

$$\epsilon^{abef} \epsilon_{abcd} = -2 \delta_{cd}^{ef} = -2(\delta_c^e \delta_d^f - \delta_d^e \delta_c^f). \quad (60)$$

Finally we use the result, established via the vielbein postulate, that the frame components of the curvature two-form are the orthonormal components of the Riemann tensor,

$$R^{cd} = \frac{1}{2} R^{cd}_{ef} e^e \wedge e^f, \quad R^{ab}_{ab} = R \quad (\text{the Ricci scalar}). \quad (61)$$

We compute $\epsilon_{abcd} e^a \wedge e^b \wedge R^{cd}$. Expand the curvature with (61),

$$\epsilon_{abcd} e^a \wedge e^b \wedge R^{cd} = \frac{1}{2} \epsilon_{abcd} R^{cd}_{ef} e^a \wedge e^b \wedge e^e \wedge e^f. \quad (62)$$

We use here the four-form identity of (59), $e^a \wedge e^b \wedge e^c \wedge e^d = -\epsilon^{abcd} \text{vol}$,

$$\epsilon_{abcd} e^a \wedge e^b \wedge R^{cd} = -\frac{1}{2} \epsilon_{abcd} \epsilon^{abef} R^{cd}_{ef} \text{vol}. \quad (63)$$

Then we contract the two symbols with (60), $\epsilon^{abef} \epsilon_{abcd} = -2(\delta_c^e \delta_d^f - \delta_d^e \delta_c^f)$,

$$-\frac{1}{2} \epsilon_{abcd} \epsilon^{abef} R^{cd}_{ef} \text{vol} = -\frac{1}{2} (-2)(\delta_c^e \delta_d^f - \delta_d^e \delta_c^f) R^{cd}_{ef} \text{vol} = (\delta_c^e \delta_d^f - \delta_d^e \delta_c^f) R^{cd}_{ef} \text{vol}. \quad (64)$$

Then, the two deltas term by term, using the antisymmetry $R^{cd}_{dc} = -R^{cd}_{cd}$ gives us,

$$\delta_c^e \delta_d^f R^{cd}_{ef} = R^{cd}_{cd}, \quad \delta_d^e \delta_c^f R^{cd}_{ef} = R^{cd}_{dc} = -R^{cd}_{cd}, \quad (65)$$

so the bracket gives $R^{cd}_{cd} - (-R^{cd}_{cd}) = 2R^{cd}_{cd}$. Finally $R^{cd}_{cd} = R^{ab}_{ab} = R$ by (61), and

$$\boxed{\epsilon_{abcd} e^a \wedge e^b \wedge R^{cd} = 2 R^{cd}_{cd} \text{vol} = 2R \text{vol} = 2R \sqrt{-g} d^4x} \quad (66)$$

Substituting $\omega = \dot{\omega}(e)$ into (58) and using the result (66),

$$S_{\text{EC}}[e, \dot{\omega}(e)] = \frac{1}{4\kappa} \int_M \epsilon_{abcd} e^a \wedge e^b \wedge R^{cd}(\dot{\omega}) = \frac{1}{4\kappa} \int_M 2R \sqrt{-g} d^4x = \frac{1}{2\kappa} \int_M \sqrt{-g} R d^4x. \quad (67)$$

With $\kappa = 8\pi G$ this is exactly the Einstein–Hilbert action,

$$\boxed{S_{\text{EC}}[e, \dot{\omega}(e)] = \frac{1}{16\pi G} \int_M \sqrt{-g} R d^4x = S_{\text{EH}}[g]} \quad (68)$$

As a conclusion we recall the procedure. The first step is a *kinematical reduction of the connection*: in Einstein–Cartan the vielbein e^a and the spin connection ω^{ab} enter as independent fields, but the connection's own field equation $\epsilon_{abcd} T^c \wedge e^d = 0$ is *algebraic* in ω so it does

not propagate the connection but fixes it, forcing $T^a = 0$ and hence $\omega = \hat{\omega}(e)$. An a priori independent connection is thereby demoted to one determined entirely by the vielbein, i.e. by the metric. The second is a *connection between actions*: we evaluate the Einstein–Cartan Lagrangian in the orthonormal frame, where the curvature two-form is already built from $\hat{\omega}(e)$, so its components $R^{ab}{}_{cd}$ are the orthonormal components of the Riemann tensor — $\hat{\omega}(e)$ being nothing but the Christoffel connection written in that frame, the two related by the vielbein postulate. The contraction $\epsilon_{abcd} e^a \wedge e^b \wedge R^{cd} = 2R \text{vol}$ then turns S_{EC} into S_{EH} .

The lesson is that two structures which are logically independent are welded together *dynamically*: metric compatibility and the vanishing of torsion emerge as *outputs* of the variational principle rather than being postulated at the outset. This is the Palatini idea also: in principle one assumes less geometric structure and recovers it on shell. In this light, Einstein–Hilbert gravity is precisely the torsion-free vacuum sector of Einstein–Cartan; the two theories become indistinguishable whenever matter carries no spin, since then nothing sources the torsion.

3. Einstein–Dirac–AdS

Problem 3.1. *The Einstein–Dirac–AdS theory in three dimensions is given by*

$$I = I_G + I_D, \quad (69)$$

where the Einstein and Dirac actions are

$$I_G = \kappa \int \epsilon_{abc} \left(R^{ab} + \frac{1}{3\ell^2} e^a \wedge e^b \right) \wedge e^c, \quad (70)$$

$$I_D = - \int \left[\epsilon_{abc} \frac{1}{4} \left(\bar{\psi} \gamma^a \nabla \psi - \nabla \bar{\psi} \gamma^a \psi \right) \wedge e^b \wedge e^c - \frac{m}{6} \bar{\psi} \psi \epsilon_{abc} e^a \wedge e^b \wedge e^c \right]. \quad (71)$$

Here, $\kappa = (16\pi G)^{-1}$, $\bar{\psi} = i\psi^\dagger \gamma^0$ is the adjoint of the Dirac spinor, and m is the mass of the Dirac spinor. Moreover, $\nabla \psi$ is the AdS covariant derivative of the Dirac field, namely,

$$\nabla \psi = d\psi + \frac{1}{2} W^{AB} J_{AB} \psi = d\psi + \frac{1}{2} \omega^{ab} J_{ab} \psi + \frac{1}{\ell} e^a P_a \psi, \quad (72)$$

where the AdS_3 generators of $SO(2,2)$ in the spinorial representation are

$$J_{ab} = \frac{1}{2} \gamma_{ab}, \quad P_a = \frac{1}{2} \gamma_a. \quad (73)$$

Here, $\gamma_{ab} = \gamma_{[a} \gamma_{b]}$, with γ_a the Dirac γ -matrices that generate the Clifford algebra $\{\gamma_a, \gamma_b\} = 2\eta_{ab} \mathbb{I}$. In $2+1$ dimensions there exist two inequivalent spinorial representations of $SO(2,1)$, labeled by $\theta = \pm 1$, that is,

$$\gamma_0 = i\theta \tau_1, \quad \gamma_1 = \theta \tau_2, \quad \gamma_2 = \theta \tau_3, \quad (74)$$

where τ_i are the Pauli matrices. Show that the gamma matrices satisfy the relation

$$\gamma_{ab} = -\theta \epsilon_{abc} \gamma^c, \quad (75)$$

and that the dual generators of $SO(2,1)$, $J_a = \frac{1}{2} \epsilon_{abc} J^{bc}$, can be written as $J_a = \frac{\theta}{2} \gamma_a = \theta P_a$. Show that the AdS covariant derivative of the Dirac spinor in Eq. (9) takes the form

$$\nabla \psi = d\psi - \frac{\theta}{2} \phi \psi + \frac{1}{2\ell} \not{\phi} \psi, \quad (76)$$

where $\phi = \omega^a \gamma_a$ and $\not\phi = e^a \gamma_a$.

Show that the field equations of the Einstein–Dirac–AdS action, obtained by performing arbitrary variations with respect to the dreibein, the Lorentz connection, and the Dirac fields, are

$$\kappa \epsilon_{abc} \left(R^{bc} + \frac{1}{\ell^2} e^b \wedge e^c \right) = \tau_a, \quad (77)$$

$$2\kappa \epsilon_{abc} T^c = \Sigma_{ab}, \quad (78)$$

$$\epsilon_{abc} \gamma^a \nabla \psi \wedge e^b \wedge e^c = \nabla \star \not\psi + \frac{m}{3} \psi \epsilon_{abc} e^a \wedge e^b \wedge e^c, \quad (79)$$

where $\star \not\phi = \frac{1}{2} \gamma^a \epsilon_{abc} e^b \wedge e^c$, while the energy–momentum tensor and the spin-density 2-forms are given, respectively, by

$$\tau_a = \frac{1}{2} \epsilon_{abc} \left(\bar{\psi} \gamma^b \nabla \psi - \overline{\nabla \psi} \gamma^b \psi - m \bar{\psi} \psi e^b \right) \wedge e^c + \frac{1}{4\ell} \bar{\psi} \psi \epsilon_{abc} e^b \wedge e^c, \quad (80)$$

$$\Sigma_{ab} = \frac{1}{8} \bar{\psi} \{ \gamma_{ab}, \gamma^c \} \psi \epsilon_{cde} e^d \wedge e^e. \quad (81)$$

Then, using the field equations, show that

$$T^a = -\frac{\theta}{8\kappa} \bar{\psi} \psi \epsilon^a_{bc} e^b \wedge e^c. \quad (82)$$

Gamma matrices and dual generators

The first relation we want to show is that the gamma matrices satisfy

$$\gamma_{ab} = -\theta \epsilon_{abc} \gamma^c. \quad (83)$$

We can show that the gamma matrices satisfy it just by replacing the indices of that matrices. We start for example with

$$\gamma_{01} = \gamma_{[0}\gamma_{1]} = \frac{1}{2} (\gamma_0 \gamma_1 - \gamma_1 \gamma_0) = \frac{1}{2} (i\theta \tau_1 \cdot \theta \tau_2 - \theta \tau_2 \cdot i\theta \tau_1) \quad (84)$$

$$= \frac{i\theta^2}{2} (\tau_1 \tau_2 - \tau_2 \tau_1) = \frac{i\theta^2}{2} [\tau_1, \tau_2], \quad (85)$$

using the commutation relation for the Pauli matrices³ and the fact that $\theta^2 = 1$, we have

$$\gamma_{01} = \frac{i\theta^2}{2} (2i\epsilon_{123} \tau_3) = -\tau_3, \quad (86)$$

and if we check the right hand side

$$-\theta \epsilon_{01c} \gamma^c = -\theta \epsilon_{012} \gamma^2 = -\theta (\theta \tau_3) = -\tau_3, \quad (87)$$

so the first case is true. We must check now

$$\gamma_{12} = \frac{1}{2} (\gamma_1 \gamma_2 - \gamma_2 \gamma_1) = \frac{1}{2} (\theta \tau_2 \cdot \theta \tau_3 - \theta \tau_3 \cdot \theta \tau_2) \quad (88)$$

$$= \frac{\theta^2}{2} (\tau_2 \tau_3 - \tau_3 \tau_2) = \frac{\theta^2}{2} [\tau_2, \tau_3] = \frac{\theta}{2} (2i\theta \epsilon_{231} \tau_1) \quad (89)$$

$$= -i\tau_1, \quad (90)$$

³The commutation relation is $[\tau_i, \tau_j] = 2i\epsilon_{ijk} \tau_k$.

and the right hand side is

$$-\theta\epsilon_{120}\gamma^0 = -\theta(i\theta\tau_1) = -i\tau_1, \quad (91)$$

so this combination also satisfy the relation. The last one is given by

$$\gamma_{02} = \frac{1}{2}(\gamma_0\gamma_2 - \gamma_2\gamma_0) = \frac{1}{2}(\theta\tau_1 \cdot i\theta\tau_3 - i\theta\tau_3 \cdot \theta\tau_1) \quad (92)$$

$$= \frac{i\theta^2}{2}[\tau_1, \tau_3] = \frac{i\theta}{2}(2i\theta\epsilon_{132}\tau_2) = \tau_2, \quad (93)$$

the right hand side of the relation for this indices gives us

$$-\theta\epsilon_{02c}\gamma^c = -\theta\epsilon_{021}\gamma^1 = \theta\theta\tau_2 = \tau_2, \quad (94)$$

so all the combinations satisfy this relation, because the other possible combinations are relative to a minus sign in the two sides of the relation, so it is directly seen that they also satisfy the relation. The dual generators of $SO(2, 1)$ are

$$J_a = \frac{1}{2}\epsilon_{abc}J^{bc}, \quad (95)$$

and we know that $J_{ab} = \frac{1}{2}\gamma_{ab}$, so we can just replace the expression. First notice that we need to raise the indices

$$J^{ab} = J_{cd}\eta^{ac}\eta^{bd} = \frac{1}{2}\eta^{ac}\eta^{bd}\gamma_{cd} = \frac{1}{2}\gamma^{ab}, \quad (96)$$

then we can see that

$$J_a = \frac{1}{2}\epsilon_{abc}J^{bc} = \frac{1}{2}\epsilon_{abc}\frac{1}{2}\gamma^{bc} = \frac{1}{4}\epsilon_{abc}\gamma^{bc}, \quad (97)$$

now we can use the previous result $\gamma_{ab} = -\theta\epsilon_{abc}\gamma^c$, then⁴

$$J_a = \frac{1}{4}\epsilon_{abc}\gamma^{bc} = -\frac{\theta}{4}(\epsilon_{abc}\epsilon_d^{bc}\gamma^d) = -\frac{\theta}{4}(2!)(\det(\eta)\eta_{ad})\gamma^d = \frac{\theta}{2}\gamma_a = \theta P_a, \quad (98)$$

where in the last line we have just used the definitions of the homework to identify the generator P_a .

Covariant derivative

We start from the definition of the covariant derivative AdS of the Dirac field

$$\nabla\psi = d\psi + \frac{1}{2}W^{AB}J_{AB}\psi = d\psi + \frac{1}{2}\omega^{ab}J_{ab}\psi + \frac{1}{l}e^aP_a\psi. \quad (99)$$

We write the generators of AdS_3 algebra in the spinor representation

$$J_{ab} = \frac{1}{2}\gamma_{ab}, P_a = \frac{1}{2}\gamma_a, \quad (100)$$

so the covariant derivative by directly replacing of this generators is

$$\nabla\psi = d\psi + \frac{1}{2}\omega^{ab}J_{ab}\psi + \frac{1}{l}e^aP_a\psi = d\psi + \frac{1}{2}\omega^{ab} \cdot \frac{1}{2}\gamma_{ab}\psi + \frac{1}{l}e^a \cdot \frac{1}{2}\gamma_a\psi. \quad (101)$$

⁴With the careful to raise the indices with the Minkowski metric η^{ab} because we have γ^{ab} , so the expression will be $\gamma^{bc} = -\theta\epsilon_d^{bc}\gamma^d$.

In order to use the notation $\psi = \omega^a \gamma_a, \not\psi = e^a \gamma_a$ we need to rewrite the expressions in term of the dual spin connection $2\omega^a = \epsilon_{abc} \omega^{bc}$, but we use again the first relation to write with $\gamma_{ab} = -\theta \epsilon_{abc} \gamma^c$

$$\frac{1}{4} \omega^{ab} \gamma_{ab} \psi = \frac{1}{4} \left(-\theta \omega^{ab} \epsilon_{abc} \gamma^c \right) = \frac{-\theta}{2} (\omega_c \gamma^c) = \frac{\theta}{2} \not\psi, \quad (102)$$

so finally we have

$$\nabla \psi = d\psi + \frac{1}{4} \omega^{ab} \gamma_{ab} \psi + \frac{1}{2l} e^a \gamma_a \psi = d\psi - \frac{\theta}{2} \not\psi + \frac{1}{2l} \not\psi. \quad (103)$$

Field equations

To obtain the Field equations of the action I we will perform the variational principle on each of the independent variables of this theory, they are the Dirac field ψ , the spin connection ω^{ab} (or the dual connection ω^a) and the dreibein e^a . We will start considering the variations w.r.t the dreibein, in particular we will take the action (70)

$$\begin{aligned} \delta_e I_G[e, \omega, \psi] &= k \delta \int \epsilon_{abc} \left(R^{ab} + \frac{1}{3l^2} e^a \wedge e^b \right) \wedge e^c \\ &= k \int \epsilon_{abc} \delta \left(R^{ab} \wedge e^c \right) + \frac{1}{3l^2} \epsilon_{abc} \delta \left(e^a \wedge e^b \wedge e^c \right) \\ &= k \int \epsilon_{abc} \left(R^{ab} + \frac{1}{l^2} e^a \wedge e^b \right) \wedge \delta e^c, \end{aligned} \quad (104)$$

where the triple dreibein term is rewriting and by renaming indices a factor 3 is out. Now the variation for the action (71)

$$\delta_e I_D[e, \omega, \psi] = \delta \int \left[\epsilon_{abc} \frac{1}{4} \left(\bar{\psi} \gamma^a \nabla \psi - \nabla \bar{\psi} \gamma^a \psi \right) \wedge e^b \wedge e^c - \frac{m}{6} \bar{\psi} \psi \epsilon_{abc} e^a \wedge e^b \wedge e^c \right], \quad (105)$$

taking each term separately for see it more neat along the calculation. Starting with

$$\begin{aligned} \delta \left(\frac{1}{4} \epsilon_{abc} \left(\bar{\psi} \gamma^a \nabla \psi - \nabla \bar{\psi} \gamma^a \psi \right) \wedge e^b \wedge e^c \right) &= \frac{1}{4} \epsilon_{abc} \delta \left(\bar{\psi} \gamma^a \nabla \psi - \nabla \bar{\psi} \gamma^a \psi \right) \wedge e^b \wedge e^c \\ &\quad + \frac{1}{4} \epsilon_{abc} \left(\bar{\psi} \gamma^a \nabla \psi - \nabla \bar{\psi} \gamma^a \psi \right) \delta \left(\wedge e^b \wedge e^c \right), \end{aligned} \quad (106)$$

here we must use a very useful properties of the covariant derivative that are explained in the box below.

Covariant derivative

Will be useful to know the form of the covariant derivative acting on the Dirac field. We start demanding that the scalar $\bar{\psi} \psi$ be a Lorentz invariant, i.e $\nabla(\bar{\psi} \psi) = d(\bar{\psi} \psi)$. With this in mind we can see how is the covariant derivative of the conjugated field, considering that the connection is written more simply as A , we have $\nabla \psi = d\psi + A\psi$, then we suppose the conjugated have the same form $\nabla \bar{\psi} = d\bar{\psi} + \alpha A \bar{\psi}$, now we determine α . The covariant derivative acts following the Leibniz rule

$$\begin{aligned} \nabla(\bar{\psi} \psi) &= \bar{\psi} \nabla \psi + \nabla \bar{\psi} \psi = \bar{\psi} (d\psi + A\psi) + (d\bar{\psi} + \alpha A \bar{\psi}) \psi \\ &= \bar{\psi} d\psi + d\bar{\psi} \psi + \bar{\psi} A \psi + \alpha A \bar{\psi} \psi = d(\bar{\psi} \psi) + \bar{\psi} (1 + \alpha) A \psi, \end{aligned} \quad (107)$$

so to be Lorenz invariant the covariant derivative must be

$$\nabla \bar{\psi} = d\bar{\psi} - A \bar{\psi}. \quad (108)$$

So with this in mind we have in the first term only contributions of the covariant derivatives, which are

$$\delta_e \nabla \psi = \delta \left(d\psi - \frac{\theta}{2} \phi \psi + \frac{1}{2l} \not{\phi} \psi \right) = \frac{1}{2l} \delta e^a \gamma_a \psi, \quad (109)$$

$$\delta_e \nabla \bar{\psi} = \delta \left(d\bar{\psi} + \frac{\theta}{2} \phi \bar{\psi} - \frac{1}{2l} \not{\phi} \bar{\psi} \right) = -\frac{1}{2l} \delta e^a \gamma_a \bar{\psi}, \quad (110)$$

so the first term of the variation that we are looking is

$$\begin{aligned} \frac{1}{4} \epsilon_{abc} \delta \left(\bar{\psi} \gamma^a \nabla \psi - \nabla \bar{\psi} \gamma^a \psi \right) \wedge e^b \wedge e^c &= \frac{1}{4} \epsilon_{abc} \delta \left(\bar{\psi} \gamma^a \left(\frac{1}{2l} \gamma_d \psi \delta e^d \right) - \left(-\frac{1}{2l} \bar{\psi} \gamma_d \delta e^d \right) \gamma^a \psi \right) \wedge e^b \wedge e^c \\ &= \frac{1}{4} \frac{1}{2l} \epsilon_{abc} \left[\bar{\psi} \gamma^a \gamma_d \psi + \bar{\psi} \gamma_d \gamma^a \psi \right] \delta e^d \wedge e^b \wedge e^c \\ &= \frac{1}{8l} \epsilon_{abc} \bar{\psi} \left[\eta^{af} \gamma_f \gamma_d + \gamma_d \eta^{af} \gamma_f \right] \psi \delta e^d \wedge e^b \wedge e^c \\ &= \frac{1}{8l} \epsilon_{abc} \bar{\psi} \eta^{af} \{ \gamma_f, \gamma_d \} \psi \delta e^d \wedge e^b \wedge e^c \\ &= \frac{1}{4l} \epsilon_{abc} \bar{\psi} \eta^{af} \eta_{fd} \psi \delta e^d \wedge e^b \wedge e^c \\ &= \frac{1}{4l} \epsilon_{abc} \bar{\psi} \psi \delta e^a \wedge e^b \wedge e^c \\ &= \frac{1}{4l} \epsilon_{abc} \bar{\psi} \psi e^a \wedge e^b \wedge \delta e^c. \end{aligned} \quad (111)$$

Where in the last part we see that by renaming indices and permutations between the wedge product and the indices or the Levi-Civita cancel all the minus signs. The second term comes directly from the variation of the dreibeins in the wedge product

$$\begin{aligned} \frac{1}{4} \epsilon_{abc} \left(\bar{\psi} \gamma^a \nabla \psi - \nabla \bar{\psi} \gamma^a \psi \right) \delta \left(\wedge e^b \wedge e^c \right) &= \frac{1}{4} \epsilon_{abc} \delta \left(\bar{\psi} \gamma^a \nabla \psi - \nabla \bar{\psi} \gamma^a \psi \right) \delta \left(\wedge e^b \wedge e^c \right) \\ &= \frac{1}{4} \epsilon_{abc} A^a \delta e^b \wedge e^c + \frac{1}{4} \epsilon_{abc} A^a e^b \wedge \delta e^c \\ &= \frac{1}{4} \epsilon_{acb} A^a \delta e^c \wedge e^b + \frac{1}{4} \epsilon_{abc} A^a e^b \wedge \delta e^c \\ &= \frac{1}{4} (-\epsilon_{abc}) A^a \left(-e^b \wedge \delta e^c \right) + \frac{1}{4} \epsilon_{abc} A^a e^b \wedge \delta e^c \\ &= \frac{1}{4} \epsilon_{abc} A^a e^b \wedge \delta e^c + \frac{1}{4} \epsilon_{abc} A^a e^b \wedge \delta e^c, \end{aligned} \quad (112)$$

where we just replaced the kinetic term by A^a for space in the lines, so now we have directly

$$\frac{1}{4} \epsilon_{abc} \left(\bar{\psi} \gamma^a \nabla \psi - \nabla \bar{\psi} \gamma^a \psi \right) \delta \left(\wedge e^b \wedge e^c \right) = \frac{1}{2} \epsilon_{abc} \left(\bar{\psi} \gamma^a \nabla \psi - \nabla \bar{\psi} \gamma^a \psi \right) e^b \wedge \delta e^c, \quad (113)$$

so finally we can add to obtain this expressions

$$\frac{1}{4} \delta \left(\epsilon_{abc} \left(\bar{\psi} \gamma^a \nabla \psi - \nabla \bar{\psi} \gamma^a \psi \right) \wedge e^b \wedge e^c \right) = \left[\frac{1}{2} \epsilon_{abc} \left(\bar{\psi} \gamma^a \nabla \psi - \nabla \bar{\psi} \gamma^a \psi \right) \wedge e^b + \frac{1}{4l} \epsilon_{abc} \bar{\psi} \psi e^a \wedge e^b \right] \wedge \delta e^c. \quad (114)$$

Now the mass term is also direct to compute because depends explicitly in the triple dreibein product

$$\begin{aligned} \frac{m}{6} \bar{\psi} \psi \epsilon_{abc} \delta \left(e^a \wedge e^b \wedge e^c \right) &= \frac{m}{6} \bar{\psi} \psi \left(\epsilon_{abc} \delta e^a \wedge e^b \wedge e^c + \epsilon_{abc} e^a \wedge \delta e^b \wedge e^c + \epsilon_{abc} e^a \wedge e^b \wedge \delta e^c \right) \\ &= \frac{m}{2} \bar{\psi} \psi \epsilon_{abc} e^a \wedge e^b \wedge \delta e^c, \end{aligned} \quad (115)$$

here we used again the signs coming from the permutations of indices in the Levi-Civita and wedge products. Now, the full variation with respect to the dreibein and asking for an extremum $\delta_e I \stackrel{!}{=} 0$ comes from adding (104), (114) and (115)

$$\begin{aligned} \delta_e I = & k \int \epsilon_{abc} \left(R^{ab} + \frac{1}{l^2} e^a \wedge e^b \right) \wedge \delta e^c \\ & - \int \left[\frac{1}{2} \epsilon_{abc} \left(\bar{\psi} \gamma^a \nabla \psi - \nabla \bar{\psi} \gamma^a \psi \right) \wedge e^b + \frac{1}{4l} \epsilon_{abc} \bar{\psi} \psi e^a \wedge e^b \right] \wedge \delta e^c - \frac{m}{2} \bar{\psi} \psi \epsilon_{abc} e^a \wedge e^b \wedge \delta e^c = 0, \end{aligned} \quad (116)$$

so we only have to factorize and we will obtain all the terms under the integral factorized by δe^c , so $\forall \delta e^c$

$$k \epsilon_{abc} \left(R^{ab} + \frac{1}{l^2} e^a \wedge e^b \right) = \frac{1}{2} \epsilon_{abc} \left[\bar{\psi} \gamma^a \nabla \psi - \nabla \bar{\psi} \gamma^a \psi - m \bar{\psi} \psi e^a \right] \wedge e^b + \frac{1}{4l} \epsilon_{abc} \bar{\psi} \psi e^a \wedge e^b, \quad (117)$$

where the term in the right hand side is exactly the energy-momentum tensor

$$\tau_c = \frac{1}{2} \epsilon_{abc} \left[\bar{\psi} \gamma^a \nabla \psi - \nabla \bar{\psi} \gamma^a \psi - m \bar{\psi} \psi e^a \right] \wedge e^b + \frac{1}{4l} \epsilon_{abc} \bar{\psi} \psi e^a \wedge e^b. \quad (118)$$

Now we compute the variation of the action with respect to the spin connection. We ask again for $\delta_\omega I \stackrel{!}{=} 0$ and we perform term by term the variation to add all at the end. The first term comes from the Einstein-Cartan term

$$\begin{aligned} \delta_\omega I_G[e, \omega, \psi] &= k \delta \int \epsilon_{abc} \left(R^{ab} + \frac{1}{3l^2} e^a \wedge e^b \right) \wedge e^c = k \int \epsilon_{abc} \delta \left(R^{ab} \right) \wedge e^c \\ &= k \int \epsilon_{abc} \delta \left(d\omega^{ab} + \omega_d^a \wedge \omega_b^d \right) \wedge e^c = k \int \epsilon_{abc} D \left(\delta\omega^{ab} \right) \wedge e^c \\ &= k \int \epsilon_{abc} \left[D \left(\delta\omega^{ab} \wedge e^c \right) + \left(\delta\omega^{ab} \wedge D e^c \right) \right] \\ &= k \int \epsilon_{abc} \delta\omega^{ab} \wedge T^c. \end{aligned} \quad (119)$$

Here D is the exterior covariant derivative, which satisfy $D(\alpha^{ab}) \wedge \beta^c = D(\alpha^{ab} \wedge \beta^c) + \alpha^{ab} \wedge D\beta^c$ and we discarded the boundary term⁵. Now we compute the variation of the Dirac spinor terms; the first of the two contributions again comes from the covariant derivatives in the kinetic term, so we use

$$\delta_\omega \nabla \psi = \delta \left(-\frac{\theta}{2} \not{\omega} \psi \right) = -\frac{\theta}{2} \delta \left(\frac{1}{2} \epsilon_{ab}^c \omega^{ab} \gamma_c \psi \right) = -\frac{\theta}{4} \left(\epsilon_{abc} \delta\omega^{ab} \gamma^c \psi \right) = \frac{1}{4} \gamma_{ab} \delta\omega^{ab} \psi, \quad (120)$$

$$\delta_\omega \nabla \bar{\psi} = \delta \left(\frac{\theta}{2} \not{\omega} \bar{\psi} \right) = -\frac{1}{4} \left(\gamma_{ab} \delta\omega^{ab} \right) \bar{\psi}. \quad (121)$$

The variation of the kinetic term is then

$$\begin{aligned} \delta_\omega \left(\frac{1}{4} \epsilon_{abc} \left(\bar{\psi} \gamma^a \nabla \psi - \nabla \bar{\psi} \gamma^a \psi \right) \wedge e^b \wedge e^c \right) &= \frac{1}{4} \epsilon_{dcf} \left(\bar{\psi} \gamma^d \left(\frac{1}{4} \gamma_{ab} \delta\omega^{ab} \right) - \left(-\frac{1}{4} \gamma_{ab} \delta\omega^{ab} \right) \gamma^d \bar{\psi} \right) \wedge e^c \wedge e^f \\ &= \frac{1}{16} \epsilon_{dcf} \bar{\psi} \{ \gamma^d, \gamma_{ab} \} \psi \delta\omega^{ab} \wedge e^c \wedge e^f, \end{aligned} \quad (122)$$

⁵The Stokes theorem is with the exterior derivative, but it is possible to show that $\epsilon_{abc} D(\delta\omega^{ab} \wedge e^c) = d(\epsilon_{abc} \delta\omega^{ab} \wedge e^c) + \epsilon_{abc} \delta\omega^{ab} \wedge T^c$.

where we can use the symmetry of the anti-commutator $\{A, B\} = \{B, A\}$ to obtain the expression of the enunciate. The next contribution comes from the mass term but it doesn't depend on the connection

$$\delta_\omega \left(\frac{m}{6} \bar{\psi} \psi \epsilon_{abc} e^a \wedge e^b \wedge e^c \right) = 0. \quad (123)$$

So the variational principle on the action varied with respect to the connection is given by the addition of

$$\delta_\omega I = k \int \epsilon_{abc} \delta \omega^{ab} \wedge T^c - \frac{1}{16} \int \epsilon_{def} \bar{\psi} \{ \gamma_{ab}, \gamma^d \} \delta \omega^{ab} \psi \wedge e^c \wedge e^f = 0 \quad (124)$$

so $\forall \delta \omega^{ab}$ we have

$$2k \epsilon_{abc} T^c = \Sigma_{ab} \quad , \quad \Sigma_{ab} = \frac{1}{8} \epsilon_{def} \bar{\psi} \{ \gamma_{ab}, \gamma^d \} \psi \wedge e^c \wedge e^f, \quad (125)$$

where Σ_{ab} is the spin density 2-form. We perform the last variation of this action which corresponds to vary with respect to the Dirac field $\bar{\psi}$ the action (in principle ψ and $\bar{\psi}$ are independent d.o.f's but we will show how it works). From the first action we have no contributions

$$\delta I_G = k \delta \int \epsilon_{abc} \left(R^{ab} + \frac{1}{3l^2} e^a \wedge e^b \right) \wedge e^c = 0. \quad (126)$$

From the kinetic and mass terms we will have non-trivial contributions. The first one is coming from

$$\epsilon_{abc} \delta \left(\bar{\psi} \gamma^a \nabla \psi - \nabla \bar{\psi} \gamma^a \psi \right) \wedge e^b \wedge e^c = \epsilon_{abc} \delta \bar{\psi} \gamma^a \nabla \psi \wedge e^b \wedge e^c - \epsilon_{abc} \delta (\nabla \bar{\psi}) \gamma^a \psi \wedge e^b \wedge e^c, \quad (127)$$

we will focus here only in the second term to factorize the variation $\delta \bar{\psi}$, to do this we can integrate by parts

$$\begin{aligned} \epsilon_{abc} \delta (\nabla \bar{\psi}) \gamma^a \psi \wedge e^b \wedge e^c &= \epsilon_{abc} \nabla \delta \bar{\psi} \gamma^a \psi \wedge e^b \wedge e^c \\ &= \epsilon_{abc} \nabla \left(\delta \bar{\psi} \gamma^a \psi \wedge e^b \wedge e^c \right) + \epsilon_{abc} \delta \bar{\psi} \gamma^a \nabla \psi \wedge e^b \wedge e^c + \epsilon_{abc} \delta \bar{\psi} \gamma^a \psi \nabla \left(\wedge e^b \wedge e^c \right), \end{aligned} \quad (128)$$

where the last term can be worked out again using the fact that $\nabla e^a = T^a$

$$\begin{aligned} \epsilon_{abc} \delta \bar{\psi} \gamma^a \psi \nabla \left(\wedge e^b \wedge e^c \right) &= \epsilon_{abc} \delta \bar{\psi} \gamma^a \psi \left(\nabla e^b \wedge e^c + e^b \wedge \nabla e^c \right) \\ &= 2 \epsilon_{abc} \delta \bar{\psi} \gamma^a \psi T^b \wedge e^c, \end{aligned} \quad (129)$$

so adding all the variation terms we can see that, from (127), the variation of the kinetic term is given by

$$\begin{aligned} -\frac{1}{4} \int \epsilon_{abc} \delta \left(\bar{\psi} \gamma^a \nabla \psi - \nabla \bar{\psi} \gamma^a \psi \right) \wedge e^b \wedge e^c &= -\frac{1}{4} \int 2 \epsilon_{abc} \delta \bar{\psi} \gamma^a \nabla \psi \wedge e^b \wedge e^c - \frac{1}{4} \int 2 \epsilon_{abc} \delta \bar{\psi} \gamma^a \psi T^b \wedge e^c + \text{B.T} \\ &= -\frac{1}{2} \int \epsilon_{abc} \delta \bar{\psi} \left(\gamma^a \nabla \psi \wedge e^b + \gamma^a \psi \wedge T^b \right) \wedge e^c. \end{aligned} \quad (130)$$

Now, from the mass term the variation is direct

$$\int \delta_{\bar{\psi}} \left(\frac{m}{6} \bar{\psi} \psi \epsilon_{abc} (e^a \wedge e^b \wedge e^c) \right) = \int \frac{m}{6} \delta \bar{\psi} \psi \epsilon_{abc} e^a \wedge e^b \wedge e^c. \quad (131)$$

With this contribution we can see that the variation of the action with respect to the Dirac field is

$$\delta_{\bar{\psi}} I_D = -\frac{1}{2} \int \epsilon_{abc} \delta \bar{\psi} \left(\gamma^a \nabla \psi \wedge e^b + \gamma^a \psi \wedge T^b \right) \wedge e^c + \int \frac{m}{6} \epsilon_{abc} \delta \bar{\psi} \psi e^a \wedge e^b \wedge e^c, \quad (132)$$

and the equation of motion $\forall \delta \bar{\psi}$ are

$$\begin{aligned} & -\frac{1}{2} \epsilon_{abc} \left(\gamma^a \nabla \psi \wedge e^b + \gamma^a \psi \wedge T^b \right) \wedge e^c + \frac{m}{6} \epsilon_{abc} \psi e^a \wedge e^b \wedge e^c = 0 \\ & \epsilon_{abc} \gamma^a \nabla \psi \wedge e^b \wedge e^c + \epsilon_{abc} \gamma^a \psi \wedge T^b \wedge e^c = \frac{m}{3} \epsilon_{abc} \psi e^a \wedge e^b \wedge e^c \\ & \epsilon_{abc} \gamma^a \nabla \psi \wedge e^b \wedge e^c = -\gamma^a \psi \wedge T^b \wedge e^c + \frac{m}{3} \psi \epsilon_{abc} e^a \wedge e^b \wedge e^c \end{aligned} \quad (133)$$

and, in order to match the result with the homework indication, we can proof that $\nabla \star \not{e}$ is

$$\nabla \star \not{e} = \frac{1}{2} \gamma^a \epsilon_{abc} \nabla (e^b \wedge e^c) = \frac{1}{2} \gamma^a \epsilon_{abc} (T^b \wedge e^c - e^b \wedge T^c) = \epsilon_{abc} \gamma^a T^b \wedge e^c, \quad (134)$$

so using reversing engineering, we can see that the actual equation of motion is

$$\epsilon_{abc} \gamma^a \nabla \psi \wedge e^b \wedge e^c = -\nabla \star \not{e} \psi + \frac{m}{3} \psi \epsilon_{abc} e^a \wedge e^b \wedge e^c. \quad (135)$$

Solving the connection field equation

As a final step we can use this equations of motion to show that the torsion can be described in terms of the vielbein and the Dirac field

$$2\kappa \epsilon_{abc} T^c = \frac{1}{8} \epsilon_{dcf} \bar{\psi} \left\{ \gamma_{ab}, \gamma^d \right\} \psi \wedge e^f, \quad (136)$$

where

$$T^a = D e^a = d e^a + \epsilon^a_{bc} \omega^b \wedge e^c \quad (137)$$

denotes the torsion two-form. To isolate the torsion, we first use the Clifford algebra identity

$$\boxed{\left\{ \gamma_{ab}, \gamma^d \right\} = 2 \left(\delta_b^d \gamma_a - \delta_a^d \gamma_b \right).} \quad (138)$$

Substituting this identity into Eq. (136) gives

$$2\kappa \epsilon_{abc} T^c = \frac{1}{8} \epsilon_{dcf} 2 \left(\delta_b^d \gamma_a - \delta_a^d \gamma_b \right) \bar{\psi} \psi^c \wedge e^f \quad (139)$$

$$= \frac{1}{4} \epsilon_{dcf} \left(\delta_b^d \bar{\psi} \gamma_a \psi - \delta_a^d \bar{\psi} \gamma_b \psi \right) e^c \wedge e^f. \quad (140)$$

Using the Kronecker deltas,

$$\delta_b^d \epsilon_{dcf} = \epsilon_{bcf}, \quad \delta_a^d \epsilon_{dcf} = \epsilon_{acf}, \quad (141)$$

one finds

$$\boxed{2\kappa \epsilon_{abc} T^c = \frac{1}{4} \left(\epsilon_{bcf} \bar{\psi} \gamma_a \psi - \epsilon_{acf} \bar{\psi} \gamma_b \psi \right) e^c \wedge e^f.} \quad (142)$$

For convenience we introduce the vector current

$$J_a := \bar{\psi} \gamma_a \psi. \quad (143)$$

The previous equation becomes

$$2\kappa\epsilon_{abc}T^c = \frac{1}{4}(\epsilon_{bcf}J_a - \epsilon_{acf}J_b)e^c \wedge e^f. \quad (144)$$

To eliminate the Levi-Civita tensor multiplying the torsion, we contract both sides with

$$\frac{1}{2}\epsilon^{abm}. \quad (145)$$

The left-hand side becomes

$$\frac{1}{2}\epsilon^{abm}(2\kappa\epsilon_{abc}T^c) = \kappa\epsilon^{abm}\epsilon_{abc}T^c \quad (146)$$

$$= 2\kappa T^m, \quad (147)$$

where we used the identity

$$\boxed{\epsilon^{abm}\epsilon_{abc} = 2\delta_c^m.} \quad (148)$$

Next, consider the first contraction on the right-hand side,

$$\epsilon^{abm}\epsilon_{bcf}J_a e^c \wedge e^f. \quad (149)$$

Using

$$\boxed{\epsilon^{abm}\epsilon_{bcf} = \delta_c^a \delta_f^m - \delta_f^a \delta_c^m,} \quad (150)$$

we obtain

$$\epsilon^{abm}\epsilon_{bcf}J_a e^c \wedge e^f = J_c e^c \wedge e^m - J_f e^m \wedge e^f. \quad (151)$$

Using the antisymmetry of the wedge product,

$$e^m \wedge e^f = -e^f \wedge e^m, \quad (152)$$

the second term becomes

$$J_f e^f \wedge e^m, \quad (153)$$

so that

$$\boxed{\epsilon^{abm}\epsilon_{bcf}J_a e^c \wedge e^f = 2J_a e^a \wedge e^m.} \quad (154)$$

Similarly,

$$\boxed{\epsilon^{abm}\epsilon_{acf} = \delta_f^b \delta_c^m - \delta_c^b \delta_f^m,} \quad (155)$$

which yields

$$\epsilon^{abm}\epsilon_{acf}J_b e^c \wedge e^f = J_f e^m \wedge e^f - J_c e^c \wedge e^m \quad (156)$$

$$= -2J_a e^a \wedge e^m. \quad (157)$$

Collecting both contributions,

$$2\kappa T^m = \frac{1}{8} (2J_a e^a \wedge e^m + 2J_a e^a \wedge e^m) \quad (158)$$

$$= \frac{1}{2} J_a e^a \wedge e^m. \quad (159)$$

Finally,

$$\boxed{T^m = \frac{1}{4\kappa} J_a e^a \wedge e^m,} \quad (160)$$

or, restoring the spinor bilinear,

$$\boxed{T^a = \frac{1}{4\kappa} (\bar{\psi} \gamma_b \psi) e^b \wedge e^a.} \quad (161)$$

This expression shows that, in Einstein–Cartan theory coupled to fermions, the torsion is not an independent propagating field. Instead, it is algebraically determined by the spin current carried by the Dirac field.

4. Solutions to Einstein–Cartán

Problem 4.1. For $d = 4$, solve the Einstein–Cartan equations in vacuum with cosmological constant, considering the following ansatz for the vielbein

$$e^0 = \sqrt{f(r)} dt, \quad e^1 = \sqrt{\frac{1}{f(r)}} dr, \quad e^2 = r d\theta, \quad e^3 = r \sqrt{\frac{1}{k}} \sin(\sqrt{k} \theta) d\phi, \quad (162)$$

where $k = 0, \pm 1$ is the Gaussian curvature. For which values of the cosmological constant and the Gaussian curvature does the obtained solution correspond to a black hole?

Solution

The structure equations of Maurer–Cartán for the torsion and curvature are

$$T^a = De^a = de^a + \omega^a_b \wedge e^b, \quad R^{ab} = d\omega^{ab} + \omega^a_c \wedge \omega^{cb}, \quad (163)$$

with $\omega^a_b = \omega^{ac}\eta_{cb}$. The first-order (Einstein–Cartan) action in $d = 4$ with cosmological constant is

$$S[e, \omega] = \frac{1}{2\kappa} \int_{\mathcal{M}} \epsilon_{abcd} \left(R^{ab} \wedge e^c \wedge e^d - \frac{\Lambda}{6} e^a \wedge e^b \wedge e^c \wedge e^d \right). \quad (164)$$

Variation of ω . Since $\delta R^{ab} = D(\delta\omega^{ab})$, an integration by parts gives

$$\delta_\omega S = -\frac{1}{2\kappa} \int \epsilon_{abcd} \delta\omega^{ab} \wedge D(e^c \wedge e^d), \quad D(e^c \wedge e^d) = T^c \wedge e^d - e^c \wedge T^d. \quad (165)$$

Antisymmetry in $[cd]$ collapses the two terms, so the connection equation is

$$\epsilon_{abcd} T^c \wedge e^d = 0 \quad \implies \quad T^a = 0. \quad (166)$$

In vacuum (no spin current) the torsion vanishes and ω^{ab} is the Levi-Civita connection.

Variation of e . Using $\delta(\epsilon_{abcd}R^{ab}e^ce^d) = 2\epsilon_{abcd}R^{ab} \wedge \delta e^c \wedge e^d$ and $\delta(\epsilon_{abcd}e^ae^be^ce^d) = 4\epsilon_{abcd}\delta e^a \wedge e^b \wedge e^c \wedge e^d$, the vielbein equation (free index a) is

$$\boxed{\epsilon_{abcd} \left(R^{bc} - \frac{\Lambda}{3} e^b \wedge e^c \right) \wedge e^d = 0} \quad (167)$$

This is the vacuum Einstein equation with Λ ; it is solved by the constant-curvature form $R^{ab} = \frac{\Lambda}{3} e^a \wedge e^b$, consistent with $R_{\mu\nu} = \Lambda g_{\mu\nu}$, $R = 4\Lambda$ in $d = 4$.

Transverse geometry

The last two legs build the 2-metric $d\theta^2 + \frac{1}{k} \sin^2(\sqrt{k}\theta) d\phi^2$, whose Gaussian curvature is exactly k .

The function $\frac{1}{\sqrt{k}} \sin(\sqrt{k}\theta)$ interpolates analytically between the three cases: $\rightarrow \sin \theta$ for $k = +1$ (sphere), $\rightarrow \theta$ for $k \rightarrow 0$ (plane), and $\rightarrow \sinh \theta$ for $k = -1$ (hyperbolic plane), using $\frac{1}{\sqrt{-1}} \sin(\sqrt{-1}\theta) = \sinh \theta$.

We record the inverse relations, used repeatedly below:

$$dt = \frac{1}{\sqrt{f}} e^0, \quad dr = \sqrt{f} e^1, \quad d\theta = \frac{1}{r} e^2, \quad d\phi = \frac{\sqrt{k}}{r \sin(\sqrt{k}\theta)} e^3. \quad (168)$$

We start computing the exterior derivative of the vielbein to use the torsion-free relation to identify the spin connection non-vanishing components.

$$de^0 = \frac{f'}{2\sqrt{f}} dr \wedge dt = \frac{f'}{2\sqrt{f}} (\sqrt{f} e^1) \wedge \left(\frac{1}{\sqrt{f}} e^0 \right) = -\frac{f'}{2\sqrt{f}} e^0 \wedge e^1, \quad (169)$$

$$de^1 = -\frac{1}{2} f^{-3/2} f' dr \wedge dr = 0, \quad (170)$$

$$de^2 = dr \wedge d\theta = (\sqrt{f} e^1) \wedge \left(\frac{1}{r} e^2 \right) = \frac{\sqrt{f}}{r} e^1 \wedge e^2. \quad (171)$$

For $e^3 = \frac{r}{\sqrt{k}} \sin(\sqrt{k}\theta) d\phi$,

$$de^3 = \left[\frac{1}{\sqrt{k}} \sin(\sqrt{k}\theta) dr + r \cos(\sqrt{k}\theta) d\theta \right] \wedge d\phi, \quad (172)$$

and inserting the inverse relations,

$$de^3 = \frac{\sqrt{f}}{r} e^1 \wedge e^3 + \frac{\sqrt{k} \cos(\sqrt{k}\theta)}{r \sin(\sqrt{k}\theta)} e^2 \wedge e^3. \quad (173)$$

Now we use (166) in the form $de^a = -\omega^a_b \wedge e^b$. The torsion-free connection is unique; solving the four equations gives us the non-vanishing contributions of the spin connection

Using our mostly-plus Minkowski signature $\eta_{ab} = \text{diag}(-1, 1, 1, 1)$, we can raise the first index $\omega^a_b = \eta^{ac} \omega_{cb}$. This gives us specific symmetry relations for the mixed indices:

- For one time-like and one space-like index ($a = 0, b = 1$):

$$\omega^0_1 = \eta^{00} \omega_{01} = -(-\omega_{10}) = \omega_{10} = \eta_{11} \omega^1_0 = \omega^1_0. \quad (174)$$

- For two space-like indices ($a = 1, b = 2$):

$$\omega^1_2 = \eta^{11}\omega_{12} = \omega_{12} = -\omega_{21} = -\eta_{22}\omega^2_1 = -\omega^2_1. \quad (175)$$

We start by evaluating Cartan's first structure equation for $a = 0$:

$$de^0 + \omega^0_1 \wedge e^1 + \omega^0_2 \wedge e^2 + \omega^0_3 \wedge e^3 = 0. \quad (176)$$

From the previous exterior derivative calculations, we know that $de^0 = \frac{f'}{2\sqrt{f}}e^1 \wedge e^0$. We can rewrite this by swapping the wedge product order:

$$de^0 = -\frac{f'}{2\sqrt{f}}e^0 \wedge e^1. \quad (177)$$

Substituting this into the structure equation:

$$-\frac{f'}{2\sqrt{f}}e^0 \wedge e^1 + \omega^0_1 \wedge e^1 + \omega^0_2 \wedge e^2 + \omega^0_3 \wedge e^3 = 0. \quad (178)$$

We can factor out the e^1 1-form on the right side for the first two terms:

$$\left(-\frac{f'}{2\sqrt{f}}e^0 + \omega^0_1\right) \wedge e^1 + \omega^0_2 \wedge e^2 + \omega^0_3 \wedge e^3 = 0. \quad (179)$$

Because the vielbein 1-forms $\{e^1, e^2, e^3\}$ form a linearly independent basis, the only way this entire expression can vanish identically without introducing torsion is if each grouped term vanishes independently. This yields the unique solution for the time-like components:

$$\omega^0_1 = \frac{f'}{2\sqrt{f}}e^0, \quad \omega^0_2 = 0, \quad \omega^0_3 = 0. \quad (180)$$

Next, we evaluate Cartan's first structure equation for $a = 2$:

$$de^2 + \omega^2_0 \wedge e^0 + \omega^2_1 \wedge e^1 + \omega^2_3 \wedge e^3 = 0. \quad (181)$$

We apply the symmetries derived earlier: $\omega^2_1 = -\omega^1_2$, and since we just found $\omega^0_2 = 0$, it follows that $\omega^2_0 = 0$.

From the exterior derivative calculations, we have $de^2 = \frac{\sqrt{f}}{r}e^1 \wedge e^2$. Swapping the order gives:

$$de^2 = -\frac{\sqrt{f}}{r}e^2 \wedge e^1. \quad (182)$$

Substituting everything into the $a = 2$ equation:

$$-\frac{\sqrt{f}}{r}e^2 \wedge e^1 - \omega^1_2 \wedge e^1 + \omega^2_3 \wedge e^3 = 0. \quad (183)$$

Factor out e^1 on the right side:

$$-\left(\frac{\sqrt{f}}{r}e^2 + \omega^1_2\right) \wedge e^1 + \omega^2_3 \wedge e^3 = 0. \quad (184)$$

Again, by the linear independence of the basis forms e^1 and e^3 , the term in the parenthesis must be equal to zero. This immediately dictates the form of ω^1_2 :

$$\omega^1_2 = -\frac{\sqrt{f}}{r}e^2. \quad (185)$$

We now evaluate Cartan's first structure equation for $a = 3$:

$$de^3 + \omega^3_0 \wedge e^0 + \omega^3_1 \wedge e^1 + \omega^3_2 \wedge e^2 = 0. \quad (186)$$

From our previous derivations, we know $\omega^3_0 = \eta^{33}\omega_{30} = \omega_{30} = -\omega_{03} = -\eta_{00}\omega^0_3 = \omega^0_3 = 0$. For the spatial components, we apply the antisymmetry relations: $\omega^3_1 = -\omega^1_3$ and $\omega^3_2 = -\omega^2_3$. Thus, the equation simplifies to:

$$de^3 - \omega^1_3 \wedge e^1 - \omega^2_3 \wedge e^2 = 0. \quad (187)$$

From the exterior derivative calculations, recalling that $e^3 = r\sqrt{\frac{1}{k}}\sin(\sqrt{k}\theta)d\phi$, we have:

$$de^3 = \frac{\sqrt{f}}{r}e^1 \wedge e^3 + \frac{\sqrt{k}\cos(\sqrt{k}\theta)}{r\sin(\sqrt{k}\theta)}e^2 \wedge e^3. \quad (188)$$

To factor out the basis forms e^1 and e^2 , we swap the order of the wedge products, introducing minus signs:

$$de^3 = -\frac{\sqrt{f}}{r}e^3 \wedge e^1 - \frac{\sqrt{k}\cos(\sqrt{k}\theta)}{r\sin(\sqrt{k}\theta)}e^3 \wedge e^2. \quad (189)$$

Substituting this into our simplified structure equation yields:

$$-\frac{\sqrt{f}}{r}e^3 \wedge e^1 - \frac{\sqrt{k}\cos(\sqrt{k}\theta)}{r\sin(\sqrt{k}\theta)}e^3 \wedge e^2 - \omega^1_3 \wedge e^1 - \omega^2_3 \wedge e^2 = 0. \quad (190)$$

Now, we group the terms proportional to e^1 and e^2 :

$$-\left(\frac{\sqrt{f}}{r}e^3 + \omega^1_3\right) \wedge e^1 - \left(\frac{\sqrt{k}\cos(\sqrt{k}\theta)}{r\sin(\sqrt{k}\theta)}e^3 + \omega^2_3\right) \wedge e^2 = 0. \quad (191)$$

Since e^1 and e^2 are linearly independent basis forms, their respective coefficients in the parentheses must vanish independently for the entire expression to be zero. This gives us the final two non-zero components of the spin connection:

$$\omega^1_3 = -\frac{\sqrt{f}}{r}e^3, \quad (192)$$

$$\omega^2_3 = -\frac{\sqrt{k}\cos(\sqrt{k}\theta)}{r\sin(\sqrt{k}\theta)}e^3 = -\frac{\sqrt{k}}{r}\cot(\sqrt{k}\theta)e^3. \quad (193)$$

So the non-vanishing components of the spin connection are :

$$\begin{aligned} \omega^{01} &= \frac{f'}{2\sqrt{f}}e^0 = \frac{f'}{2}dt, & \omega^{12} &= -\frac{\sqrt{f}}{r}e^2 = -\sqrt{f}d\theta, \\ \omega^{13} &= -\frac{\sqrt{f}}{r}e^3 = -\frac{\sqrt{f}}{\sqrt{k}}\sin(\sqrt{k}\theta)d\phi, & \omega^{23} &= -\frac{\sqrt{k}\cos(\sqrt{k}\theta)}{r\sin(\sqrt{k}\theta)}e^3 = -\cos(\sqrt{k}\theta)d\phi, \end{aligned} \quad (194)$$

with $\omega^{02} = \omega^{03} = 0$. We verify each equation directly, recalling $\omega^a_0 = -\omega^{a0}$ and $\omega^a_i = \omega^{ai}$:

$$-\omega^0_b \wedge e^b = -\omega^{01} \wedge e^1 = -\frac{f'}{2\sqrt{f}} e^0 \wedge e^1 = de^0, \quad (195)$$

$$-\omega^1_b \wedge e^b = -\omega^{01} \wedge e^0 - \omega^{12} \wedge e^2 - \omega^{13} \wedge e^3 = 0 = de^1, \quad (196)$$

$$-\omega^2_b \wedge e^b = -\omega^{21} \wedge e^1 - \omega^{23} \wedge e^3 = \frac{\sqrt{f}}{r} e^1 \wedge e^2 = de^2, \quad (197)$$

$$-\omega^3_b \wedge e^b = -\omega^{31} \wedge e^1 - \omega^{32} \wedge e^2 = \frac{\sqrt{f}}{r} e^1 \wedge e^3 + \frac{\sqrt{k} \cos(\sqrt{k}\theta)}{r \sin(\sqrt{k}\theta)} e^2 \wedge e^3 = de^3. \quad (198)$$

Derivation of the Curvature 2-forms R^{ab}

To compute the curvature 2-forms, we use Cartan's second structure equation:

$$R^a_b = d\omega^a_b + \omega^a_c \wedge \omega^c_b. \quad (199)$$

Since the Einstein-Cartan field equations are often expressed with both indices raised, it is convenient to raise the second index immediately. Using the tangent space metric η^{cb} , we define $\omega^{ab} = \omega^a_c \eta^{cb}$ and $R^{ab} = R^a_c \eta^{cb}$. Cartan's second structure equation with fully raised indices becomes:

$$R^{ab} = d\omega^{ab} + \omega^a_c \wedge \omega^{cb}. \quad (200)$$

Note that both ω^{ab} and R^{ab} are antisymmetric under the exchange of their indices ($R^{ab} = -R^{ba}$).

Before differentiating, let us express the non-zero spin connection 1-forms with raised indices in terms of the coordinate basis. Given $\eta^{ab} = \text{diag}(-1, 1, 1, 1)$, raising spatial indices does not change the sign, so $\omega^{12} = \omega^1_2$, $\omega^{13} = \omega^1_3$, and $\omega^{23} = \omega^2_3$. For the time-like component, $\omega^{01} = \omega^0_1 \eta^{11} = \omega^0_1$. Substituting the vielbeins, we have:

$$\omega^{01} = \frac{f'}{2} dt, \quad (201)$$

$$\omega^{12} = -\sqrt{f} d\theta, \quad (202)$$

$$\omega^{13} = -\frac{\sqrt{f}}{\sqrt{k}} \sin(\sqrt{k}\theta) d\phi, \quad (203)$$

$$\omega^{23} = -\cos(\sqrt{k}\theta) d\phi. \quad (204)$$

We start computing R^{01} with the results that we have from the previous computations

$$R^{01} = d\omega^{01} + \omega^0_c \wedge \omega^{c1}. \quad (205)$$

Since $\omega^{02} = \omega^{03} = 0$, the wedge product term vanishes. We simply take the exterior derivative:

$$R^{01} = d\left(\frac{f'}{2} dt\right) = \frac{f''}{2} dr \wedge dt. \quad (206)$$

Substituting the basis forms:

$$R^{01} = \frac{f''}{2} (\sqrt{f} e^1) \wedge \left(\frac{1}{\sqrt{f}} e^0\right) = \frac{f''}{2} e^1 \wedge e^0 = -\frac{f''}{2} e^0 \wedge e^1. \quad (207)$$

For R^{02} , the differential term $d\omega^{02}$ is zero. The only surviving wedge product term is for $c = 1$:

$$R^{02} = \omega^0_1 \wedge \omega^{12} = \left(\frac{f'}{2}dt\right) \wedge \left(-\sqrt{f}d\theta\right) = -\frac{f'\sqrt{f}}{2}dt \wedge d\theta. \quad (208)$$

Substituting the basis forms:

$$R^{02} = -\frac{f'\sqrt{f}}{2} \left(\frac{1}{\sqrt{f}}e^0\right) \wedge \left(\frac{1}{r}e^2\right) = -\frac{f'}{2r}e^0 \wedge e^2. \quad (209)$$

Similarly for R^{03} :

$$R^{03} = \omega^0_1 \wedge \omega^{13} = \left(\frac{f'}{2}dt\right) \wedge \left(-\frac{\sqrt{f}}{\sqrt{k}}\sin(\sqrt{k}\theta)d\phi\right) = -\frac{f'\sqrt{f}}{2\sqrt{k}}\sin(\sqrt{k}\theta)dt \wedge d\phi. \quad (210)$$

Substituting the basis forms:

$$R^{03} = -\frac{f'\sqrt{f}}{2\sqrt{k}}\sin(\sqrt{k}\theta) \left(\frac{1}{\sqrt{f}}e^0\right) \wedge \left(\frac{\sqrt{k}}{r\sin(\sqrt{k}\theta)}e^3\right) = -\frac{f'}{2r}e^0 \wedge e^3. \quad (211)$$

For the curvature R^{12} component we have :

$$R^{12} = d\omega^{12} + \omega^1_0 \wedge \omega^{02} + \omega^1_3 \wedge \omega^{32}. \quad (212)$$

We know $\omega^{02} = 0$. We evaluate the remaining terms. The derivative is:

$$d\omega^{12} = d(-\sqrt{f}d\theta) = -\frac{f'}{2\sqrt{f}}dr \wedge d\theta. \quad (213)$$

The wedge product term is:

$$\omega^1_3 \wedge \omega^{32} = \omega^{13} \wedge (-\omega^{23}) = \left(-\frac{\sqrt{f}}{\sqrt{k}}\sin(\sqrt{k}\theta)d\phi\right) \wedge \left(\cos(\sqrt{k}\theta)d\phi\right) = 0. \quad (214)$$

Since $d\phi \wedge d\phi = 0$, this term vanishes. Therefore:

$$R^{12} = -\frac{f'}{2\sqrt{f}}(\sqrt{f}e^1) \wedge \left(\frac{1}{r}e^2\right) = -\frac{f'}{2r}e^1 \wedge e^2. \quad (215)$$

Now, for R^{13} we have :

$$R^{13} = d\omega^{13} + \omega^1_c \wedge \omega^{c3} = d\omega^{13} + \omega^1_2 \wedge \omega^{23}. \quad (216)$$

First, we take the exterior derivative of ω^{13} :

$$d\omega^{13} = d\left(-\frac{\sqrt{f}}{\sqrt{k}}\sin(\sqrt{k}\theta)d\phi\right) = -\frac{f'}{2\sqrt{f}\sqrt{k}}\sin(\sqrt{k}\theta)dr \wedge d\phi - \sqrt{f}\cos(\sqrt{k}\theta)d\theta \wedge d\phi. \quad (217)$$

Next, we compute the wedge product term ($\omega^1_2 = \omega^{12}$):

$$\omega^1_2 \wedge \omega^{23} = (-\sqrt{f}d\theta) \wedge (-\cos(\sqrt{k}\theta)d\phi) = \sqrt{f}\cos(\sqrt{k}\theta)d\theta \wedge d\phi. \quad (218)$$

Adding these together, the terms containing $d\theta \wedge d\phi$ cancel out exactly:

$$R^{13} = -\frac{f'}{2\sqrt{f}\sqrt{k}}\sin(\sqrt{k}\theta)dr \wedge d\phi. \quad (219)$$

Substituting the basis forms:

$$R^{13} = -\frac{f'}{2\sqrt{f}\sqrt{k}} \sin(\sqrt{k}\theta) (\sqrt{f}e^1) \wedge \left(\frac{\sqrt{k}}{r \sin(\sqrt{k}\theta)} e^3 \right) = -\frac{f'}{2r} e^1 \wedge e^3. \quad (220)$$

The last expression is R^{23} :

$$R^{23} = d\omega^{23} + \omega_c^2 \wedge \omega^{c3} = d\omega^{23} + \omega_1^2 \wedge \omega^{13}. \quad (221)$$

Taking the exterior derivative of ω^{23} :

$$d\omega^{23} = d(-\cos(\sqrt{k}\theta)d\phi) = \sqrt{k} \sin(\sqrt{k}\theta) d\theta \wedge d\phi. \quad (222)$$

For the wedge product, note that $\omega_1^2 = -\omega_2^1 = \sqrt{f}d\theta$:

$$\omega_1^2 \wedge \omega^{13} = (\sqrt{f}d\theta) \wedge \left(-\frac{\sqrt{f}}{\sqrt{k}} \sin(\sqrt{k}\theta) d\phi \right) = -\frac{f}{\sqrt{k}} \sin(\sqrt{k}\theta) d\theta \wedge d\phi. \quad (223)$$

Adding the components together:

$$R^{23} = \left(\sqrt{k} - \frac{f}{\sqrt{k}} \right) \sin(\sqrt{k}\theta) d\theta \wedge d\phi = \frac{k-f}{\sqrt{k}} \sin(\sqrt{k}\theta) d\theta \wedge d\phi. \quad (224)$$

Substituting the basis forms:

$$R^{23} = \frac{k-f}{\sqrt{k}} \sin(\sqrt{k}\theta) \left(\frac{1}{r} e^2 \right) \wedge \left(\frac{\sqrt{k}}{r \sin(\sqrt{k}\theta)} e^3 \right) = \frac{k-f}{r^2} e^2 \wedge e^3. \quad (225)$$

Now we solve the vacuum field equations with a cosmological constant Λ in the 4D first-order formalism (167)

$$\epsilon_{abcd} \left(R^{bc} - \frac{\Lambda}{3} e^b \wedge e^c \right) \wedge e^d = 0 \quad (226)$$

By choosing specific values for the free index a , we can expand this totally antisymmetric equation into our independent ordinary differential equations.

First we set $a = 0$, the indices b, c, d must cycle through permutations of 1, 2, 3. The equation simplifies to:

$$R^{12} \wedge e^3 + R^{23} \wedge e^1 + R^{31} \wedge e^2 = \frac{\Lambda}{3} (e^1 \wedge e^2 \wedge e^3 + e^2 \wedge e^3 \wedge e^1 + e^3 \wedge e^1 \wedge e^2) \quad (227)$$

The right side simplifies to $\Lambda e^1 \wedge e^2 \wedge e^3$. For the left side, we substitute our derived curvature 2-forms, making sure to swap wedge products to match the volume form $e^1 \wedge e^2 \wedge e^3$:

$$R^{12} \wedge e^3 = \left(-\frac{f'}{2r} e^1 \wedge e^2 \right) \wedge e^3 = -\frac{f'}{2r} e^1 \wedge e^2 \wedge e^3 \quad (228)$$

$$R^{23} \wedge e^1 = \left(\frac{k-f}{r^2} e^2 \wedge e^3 \right) \wedge e^1 = \frac{k-f}{r^2} e^1 \wedge e^2 \wedge e^3 \quad (229)$$

$$R^{31} \wedge e^2 = -R^{13} \wedge e^2 = -\left(-\frac{f'}{2r} e^1 \wedge e^3 \right) \wedge e^2 = -\frac{f'}{2r} e^1 \wedge e^2 \wedge e^3 \quad (230)$$

Adding these contributions together gives:

$$\left(-\frac{f'}{r} + \frac{k-f}{r^2} \right) e^1 \wedge e^2 \wedge e^3 = \Lambda e^1 \wedge e^2 \wedge e^3 \quad (231)$$

Since the basis forms are linearly independent, we obtain the differential equation for $f(r)$:

$$\frac{k-f}{r^2} - \frac{f'}{r} = \Lambda \implies k - f - rf' = \Lambda r^2 \quad (232)$$

Recognizing that $f + rf' = (rf)'$, we can rewrite this as:

$$(rf)' = k - \Lambda r^2 \quad (233)$$

Integrating both sides with respect to r :

$$rf(r) = kr - \frac{\Lambda}{3}r^3 - 2M \quad (234)$$

where $-2M$ is an integration constant identified with the mass parameter (to match with the classical solution of Schwarzschild). Dividing by r yields the general metric function:

$$f(r) = k - \frac{2M}{r} - \frac{\Lambda}{3}r^2 \quad (235)$$

To this point looks like we just have the function to replace in the metric as a solution, but to ensure rigor, we must check that other components do not overconstrain the system. Setting $a = 2$, the indices cycle through 0, 1, 3:

$$R^{01} \wedge e^3 + R^{13} \wedge e^0 + R^{30} \wedge e^1 = \Lambda e^0 \wedge e^1 \wedge e^3 \quad (236)$$

Evaluating each term and swapping indices to match $e^0 \wedge e^1 \wedge e^3$:

$$R^{01} \wedge e^3 = \left(-\frac{f''}{2} e^0 \wedge e^1 \right) \wedge e^3 = -\frac{f''}{2} e^0 \wedge e^1 \wedge e^3 \quad (237)$$

$$R^{13} \wedge e^0 = \left(-\frac{f'}{2r} e^1 \wedge e^3 \right) \wedge e^0 = -\frac{f'}{2r} e^0 \wedge e^1 \wedge e^3 \quad (238)$$

$$R^{30} \wedge e^1 = -R^{03} \wedge e^1 = -\left(-\frac{f'}{2r} e^0 \wedge e^3 \right) \wedge e^1 = -\frac{f'}{2r} e^0 \wedge e^1 \wedge e^3 \quad (239)$$

Summing these gives a second order differential equation:

$$-\frac{f''}{2} - \frac{f'}{r} = \Lambda \quad (240)$$

Let us test our solution (235) against this equation. Taking the derivatives of $f(r)$:

$$f'(r) = \frac{2M}{r^2} - \frac{2\Lambda}{3}r \quad (241)$$

$$f''(r) = -\frac{4M}{r^3} - \frac{2\Lambda}{3} \quad (242)$$

Substituting these into the left side of the second order equation:

$$-\frac{1}{2} \left(-\frac{4M}{r^3} - \frac{2\Lambda}{3} \right) - \frac{1}{r} \left(\frac{2M}{r^2} - \frac{2\Lambda}{3}r \right) = \frac{2M}{r^3} + \frac{\Lambda}{3} - \frac{2M}{r^3} + \frac{2\Lambda}{3} = \Lambda \quad (243)$$

The identity is perfectly satisfied, confirming that our exact solution for $f(r)$ is consistent with the other field equation components.

Black Hole Conditions

For the obtained solution to correspond to a black hole, the spacetime must contain an event horizon. An event horizon is located at the coordinate $r_h > 0$ where the metric function vanishes, meaning $f(r_h) = 0$. We require a physical mass so we consider as assumption that $M > 0$ ⁶.

$$f(r) = k - \frac{2M}{r} - \frac{\Lambda}{3}r^2 = 0 \implies \frac{\Lambda}{3}r^3 - kr + 2M = 0 \quad (244)$$

We define the polynomial $P(r) = \frac{\Lambda}{3}r^3 - kr + 2M$. We analyze the roots depending on the cosmological constant:

- **Case $\Lambda = 0$ (Asymptotically Flat):**

The function reduces to $f(r) = k - \frac{2M}{r} = 0$, yielding a root at $r_h = \frac{2M}{k}$. Since $M > 0$ and the radial coordinate r must be positive, this configuration requires strictly $k = 1$. This is the standard spherical Schwarzschild black hole.

- **Case $\Lambda > 0$ (de Sitter):**

The polynomial whose roots define the horizons is given by:

$$P(r) = \frac{\Lambda}{3}r^3 - kr + 2M \quad (245)$$

We need to find if there is at least one positive real root $r_h > 0$ such that $P(r_h) = 0$.

First, we analyze the behavior of the polynomial at the boundaries of our physical domain ($r > 0$). At the origin, we have $P(0) = 2M > 0$. As $r \rightarrow \infty$, the cubic term dominates, so $P(r) \rightarrow +\infty$.

Because the function starts positive and ends positive, the only way it can cross the r -axis to create a root is if the function has a local minimum that drops to zero or below zero. To find the extrema, we take the first derivative with respect to r and set it to zero:

$$P'(r) = \Lambda r^2 - k = 0 \implies r^2 = \frac{k}{\Lambda} \quad (246)$$

We can immediately rule out two of the possible topological curvatures:

- If $k = 0$, $P'(r) = \Lambda r^2 > 0$ for all $r > 0$. The function is strictly increasing, so it never crosses zero.
- If $k = -1$, $P'(r) = \Lambda r^2 + 1 > 0$ for all $r > 0$. Again, the function is strictly increasing.

Therefore, to have a real, strictly positive critical point, we must have $k = 1$. The critical points are at $r = \pm \frac{1}{\sqrt{\Lambda}}$. We select the physically relevant positive coordinate:

$$r_{min} = \frac{1}{\sqrt{\Lambda}} \quad (247)$$

To formally verify this is a minimum, we evaluate the second derivative:

$$P''(r) = 2\Lambda r \implies P''(r_{min}) = 2\Lambda \left(\frac{1}{\sqrt{\Lambda}} \right) = 2\sqrt{\Lambda} > 0 \quad (248)$$

⁶The polynomial $p(r)$ determines *where* an event horizon is located, but the *geometry* of the horizon is dictated by the angular part of the vielbein ansatz. An event horizon is a 2-dimensional spatial cross-section defined at constant time ($dt = 0 \implies e^0 = 0$) and constant radius ($dr = 0 \implies e^1 = 0$).

Since $P''(r_{min}) > 0$, it is indeed a local minimum.

For a black hole to exist, the polynomial's value at this minimum must be less than or equal to zero ($P(r_{min}) \leq 0$). If $P(r_{min}) = 0$, we have an extreme black hole with a single degenerate horizon. If $P(r_{min}) < 0$, we have two distinct horizons (an event horizon and a cosmological horizon). Let us enforce this condition:

$$P(r_{min}) = \frac{\Lambda}{3}(r_{min})^3 - (1)r_{min} + 2M \leq 0 \quad (249)$$

Substituting $r_{min} = \frac{1}{\sqrt{\Lambda}}$:

$$\frac{\Lambda}{3} \left(\frac{1}{\sqrt{\Lambda}} \right)^3 - \left(\frac{1}{\sqrt{\Lambda}} \right) + 2M \leq 0 \quad (250)$$

$$\frac{\Lambda}{3\Lambda\sqrt{\Lambda}} - \frac{1}{\sqrt{\Lambda}} + 2M \leq 0 \quad (251)$$

$$\frac{1}{3\sqrt{\Lambda}} - \frac{1}{\sqrt{\Lambda}} + 2M \leq 0 \quad (252)$$

To combine the terms, we find a common denominator:

$$\left(\frac{1}{3} - 1 \right) \frac{1}{\sqrt{\Lambda}} + 2M \leq 0 \quad (253)$$

$$-\frac{2}{3\sqrt{\Lambda}} + 2M \leq 0 \quad (254)$$

$$2M \leq \frac{2}{3\sqrt{\Lambda}} \quad (255)$$

Dividing by 2 gives the final upper bound on the mass:

$$M \leq \frac{1}{3\sqrt{\Lambda}} \quad (256)$$

Thus, for $\Lambda > 0$, the spacetime represents a black hole strictly when $k = 1$.

- **$\Lambda < 0$ (asymptotically AdS).** Let us write $\Lambda = -|\Lambda|$. The polynomial becomes:

$$P(r) = \frac{|\Lambda|}{3}r^3 + kr - 2M \quad (257)$$

Evaluating the boundaries, we see that $P(0) = -2M < 0$, and as $r \rightarrow +\infty$, the cubic term dominates so $P(r) \rightarrow +\infty$. Because $P(r)$ is continuous and transitions from a negative value to positive infinity, it must cross the axis. Furthermore, it strictly increases for large r , guaranteeing there is always exactly one positive root $r_h > 0$, valid for every Gaussian curvature $k = 0, \pm 1$.

To understand the physical nature of these solutions, we examine the geometry of the event horizon. The induced metric on this 2-dimensional surface at constant time ($dt = 0$) and constant radius ($dr = 0$), denoted $d\Sigma_k^2$, is constructed from the spatial vielbeins e^2 and e^3 . Factoring out the radius r , the base space metric is:

$$d\Sigma_k^2 = \frac{1}{r^2} [(e^2)^2 + (e^3)^2] = d\theta^2 + \frac{\sin^2(\sqrt{k}\theta)}{k} d\phi^2 \quad (258)$$

By evaluating this metric for each value of k , we obtain the corresponding horizon topologies:

- $k = +1$ (**Spherical Horizon**): The metric becomes $d\Sigma_1^2 = d\theta^2 + \sin^2 \theta d\phi^2$, which is the standard line element of a 2-sphere (S^2). This yields the Schwarzschild–AdS black hole.
- $k = 0$ (**Planar/Toroidal Horizon**): Taking the limit as $k \rightarrow 0$, the function $\frac{\sin(\sqrt{k}\theta)}{\sqrt{k}} \rightarrow \theta$. The metric becomes $d\Sigma_0^2 = d\theta^2 + \theta^2 d\phi^2$, the flat metric of a 2-dimensional plane ($\mathbb{R}^2$) in polar coordinates. Through discrete coordinate identifications, this plane can be compactified into a torus (T^2), leading to planar or toroidal black branes.
- $k = -1$ (**Hyperbolic Horizon**): Using the identity $\sin(i\theta) = i \sinh(\theta)$, the metric becomes $d\Sigma_{-1}^2 = d\theta^2 + \sinh^2 \theta d\phi^2$, the standard line element of a 2-dimensional hyperbolic space (H^2). Proper discrete quotients of this space yield compact topological surfaces with genus $g \geq 2$, forming a hyperbolic (topological) black hole.

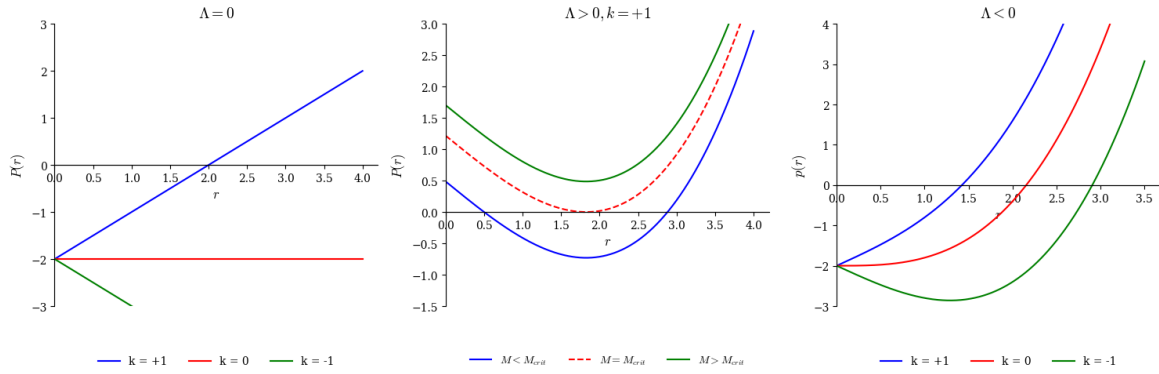


Figure 1: Polynomials determining the radial location of event horizons (r_h) via their positive roots. **Left:** In flat spacetime ($\Lambda = 0$), an event horizon forms exclusively for the spherical topology ($k = +1$). **Center:** In de Sitter space ($\Lambda > 0$ with $k = +1$), the existence of horizons is constrained by the mass bound (M_{crit}), yielding two distinct horizons ($M < M_{crit}$), one degenerate horizon ($M = M_{crit}$), or a naked singularity ($M > M_{crit}$). **Right:** In Anti-de Sitter space ($\Lambda < 0$), the dominant positive cubic term guarantees exactly one horizon for all spatial topologies ($k = +1, 0, -1$).

For all three topological cases, the singularity is successfully censored: for $r > r_h$, we have $f(r) > 0$ (the static exterior AdS region), and for $r < r_h$, $f(r) < 0$ (the dynamic black hole interior)⁷.

5. Carroll gravity

Problem 5.1. *A Carrollian gravity possesses two connections, which can be obtained algebraically from the following equations*

$$\partial_{[\mu} \tau_{\nu]} - \omega_{[\mu}{}^a e_{\nu]a} = 0, \quad (259)$$

$$\partial_{[\mu} e_{\nu]}{}^a - \omega_{[\mu}{}^{ab} e_{\nu]b} = 0. \quad (260)$$

Considering that

$$e_\mu{}^a e^\mu{}_b = \delta^a{}_b, \quad \tau^\mu \tau_\mu = 1, \quad \tau_\mu e^\mu{}_a = 0, \quad \tau^\mu e_{a\mu} = 0, \quad e^a{}_\mu e^\nu{}_a + \tau^\mu \tau_\nu = \delta^\mu{}_\nu, \quad (261)$$

⁷The code for the plot is in (A).

show that

$$\omega_\mu^a = \tau_\mu \tau^\nu e^{\rho a} \partial_{[\nu} \tau_{\rho]} + e^{\nu a} \partial_{[\mu} \tau_{\nu]} + S^{ab} e_{\mu b}, \quad (262)$$

$$\omega_\mu^{ab} = -2 e^{\rho[a} \partial_{[\mu} e_{\rho]}^{b]} + e_{\mu c} e^{\rho a} e^{\nu b} \partial_{[\rho} e_{\nu]}^c, \quad (263)$$

where $S^{ab} = S^{(ab)} = e^{\mu(a} \omega_\mu^{b)}$ corresponds to an undetermined part of the connection.

Solution for ω_μ^a

We start with the equation (259). We expand it and project all the equation with other vielbein

$$\begin{aligned} \partial_{[\mu} \tau_{\nu]} - \omega_{[\mu}^a e_{\nu]a} &= 0 \\ \omega_\mu^b e_{\nu b} - \omega_\nu^b e_{\mu b} &= 2 \partial_{[\mu} \tau_{\nu]} / \cdot e^{\nu a} \\ \omega_\mu^b \delta_b^a - \omega_\nu^b e_{b\mu} e^{\nu a} &= 2 \partial_{[\mu} \tau_{\nu]} e^{\nu a} \\ \implies \omega_\mu^a &= 2 \partial_{[\mu} \tau_{\nu]} e^{\nu a} + \omega_\nu^b e^{\nu a} e_{b\mu}, \end{aligned} \quad (264)$$

here we decompose the term $\omega_\nu^b e^{\nu a}$ into a symmetric and anti symmetric part

$$\omega_\nu^b e^{\nu a} = \omega_\nu^{[b} e^{a]\nu} + \omega_\nu^{(b} e^{a)\nu}, \quad (265)$$

and we will call rename the symmetric part as $S^{ab} \equiv \omega_\nu^{(b} e^{a)\nu}$. To determine completely this term we can take the equation (264) and project again with another vielbein

$$\begin{aligned} \omega_\mu^a &= 2 \partial_{[\mu} \tau_{\nu]} e^{\nu a} + \omega_\nu^b e^{\nu a} e_{b\mu} / \cdot e^{c\mu} \\ \omega_\mu^a e^{c\mu} &= 2 \partial_{[\mu} \tau_{\nu]} e^{\nu a} e^{c\mu} + \omega_\nu^b e^{\nu a} \delta_b^c \\ \omega_\mu^a e^{c\mu} - \omega_\mu^c e^{a\mu} &= 2 \partial_{[\mu} \tau_{\nu]} e^{\nu a} e^{c\mu}, \end{aligned} \quad (266)$$

from this result we can see that the anti symmetric part of the expression (265) obtained, the symmetric part is canceled automatically (this fact comes from the nature of the anti symmetry in the coordinate indices) so it is undetermined by this equation. Then we can write the expression

$$\omega_\nu^a e^{\nu b} = S^{(ab)} + e^{\nu b} e^{\rho a} \partial_{[\nu} \tau_{\rho]}, \quad (267)$$

and replacing this into the original equation with we started

$$\begin{aligned} \omega_\mu^a &= 2 \partial_{[\mu} \tau_{\nu]} e^{\nu a} + \omega_\nu^b e^{\nu a} e_{b\mu} \\ &= 2 \partial_{[\mu} \tau_{\nu]} e^{\nu a} + S^{(ba)} e_{b\mu} + e^{\nu a} e^{\rho b} \partial_{[\nu} \tau_{\rho]} e_{b\mu} \\ &= 2 \partial_{[\mu} \tau_{\nu]} e^{\nu a} + S^{(ba)} e_{b\mu} + e^{\nu a} \partial_{[\nu} \tau_{\rho]} \left(\delta_\mu^\rho - \tau^\rho \tau_\mu \right) \\ &= 2 \partial_{[\mu} \tau_{\nu]} e^{\nu a} + S^{(ba)} e_{b\mu} - e^{\nu a} \partial_{[\mu} \tau_{\nu]} - e^{\nu a} \partial_{[\nu} \tau_{\rho]} \tau^\rho \tau_\mu \\ &= \partial_{[\mu} \tau_{\nu]} e^{\nu a} + S^{(ba)} e_{b\mu} - e^{\nu a} \partial_{[\nu} \tau_{\rho]} \tau^\rho \tau_\mu \\ &= \partial_{[\mu} \tau_{\nu]} e^{\nu a} + S^{(ba)} e_{b\mu} + e^{\rho a} \partial_{[\nu} \tau_{\rho]} \tau^\nu \tau_\mu, \end{aligned} \quad (268)$$

where in the last step we have changed the indices $\nu \leftrightarrow \rho$ and used the anti-symmetry in the partial derivative to obtain the expression we want. To show the second one, we proceed in similar way

$$\begin{aligned} \partial_{[\mu} e_{\nu]}^a - \omega_{[\mu}^{ab} e_{\nu]b} &= 0 \\ \omega_\mu^{ab} e_{\nu b} - \omega_\nu^{ab} e_{\mu b} &= 2 \partial_{[\mu} e_{\nu]}^a / \cdot e^{\nu c} \\ \omega_\mu^{ac} - \omega_\nu^{ab} e_{\mu b} e^{\nu c} &= 2 \partial_{[\mu} e_{\nu]}^a e^{\nu c} \\ \implies \omega_\mu^{ac} &= 2 \partial_{[\mu} e_{\nu]}^a e^{\nu c} + \omega_\nu^{ab} e^{\nu c} e_{\mu b}, \end{aligned} \quad (269)$$

where we need to work out the last term which still involves the connection. In order to find that expression in terms of other quantities, we use the fact that the spin connection is anti-symmetric by definition, so we can make that explicit into our equation

$$\begin{aligned}
\omega_\mu{}^{ac} &= 2\partial_{[\mu}e_{\nu]}^ae^{\nu c} + \omega_\nu{}^{ab}e^{\nu c}e_{\mu b} \\
&= \left(\partial_{[\mu}e_{\nu]}^ae^{\nu c} - \partial_{[\mu}e_{\nu]}^ce^{\nu a}\right) + \frac{1}{2}\left(\omega_\nu{}^{ab}e^{\nu c} - \omega_\nu{}^{cb}e^{\nu a}\right)e_{\mu b} \\
&= 2\partial_{[\mu}e_{\nu]}^{[a}e^{c]\nu} + \frac{1}{2}\left(\omega_\nu{}^{ab}e^{\nu c} - \omega_\nu{}^{cb}e^{\nu a}\right)e_{\mu b}.
\end{aligned} \tag{270}$$

For the last two terms we can use the equation (260) to isolate the anti-symmetric term in parenthesis

$$\begin{aligned}
\partial_{[\rho}e_{\nu]}^be^{\nu c} &= \frac{1}{2}\left(\omega_\rho{}^{bd}e_{\nu d} - \omega_\nu{}^{bd}e_{\rho d}\right) / \cdot e^{\rho a}e^{\nu c} \\
\partial_{[\rho}e_{\nu]}^be^{\rho a}e^{\nu c} &= \frac{1}{2}\left(\omega_\rho{}^{bd}e^{\rho a}\delta_d^c - \omega_\nu{}^{bd}\delta_d^ae^{\nu c}\right) \\
&= \frac{1}{2}\left(\omega_\rho{}^{bc}e^{\rho a} - \omega_\nu{}^{ba}e^{\nu c}\right) \\
&= \frac{1}{2}\left(\omega_\nu{}^{ab}e^{\nu c} - \omega_\nu{}^{cb}e^{\nu a}\right),
\end{aligned} \tag{271}$$

so we now can simply replace the term into the total expression for the connection

$$\begin{aligned}
\omega_\mu{}^{ac} &= 2\partial_{[\mu}e_{\nu]}^{[a}e^{c]\nu} + \frac{1}{2}\left(\omega_\nu{}^{ab}e^{\nu c} - \omega_\nu{}^{cb}e^{\nu a}\right)e_{\mu b} \\
&= 2\partial_{[\mu}e_{\nu]}^{[a}e^{c]\nu} + \frac{1}{2} \cdot 2e^{\rho a}e^{\nu c}\partial_{[\rho}e_{\nu]}^be_{\mu b} \\
&= 2\partial_{[\mu}e_{\nu]}^{[a}e^{c]\nu} + e_{\mu b}e^{\rho a}e^{\nu c}\partial_{[\rho}e_{\nu]}^b,
\end{aligned} \tag{272}$$

and finally using the relation (showing where the anti-symmetric bracket is placed)

$$\begin{aligned}
2\partial_{[\mu}e_{\nu]}^{[a}e^{c]\nu} &= \partial_{[\mu}e_{\nu]}^ae^{\nu c} - \partial_{[\mu}e_{\nu]}^ce^{\nu a} \\
&= e^{\nu c}\partial_{[\mu}e_{\nu]}^a - e^{\nu a}\partial_{[\mu}e_{\nu]}^c \\
&= -2e^{\rho[a}\partial_{[\mu}e_{\rho]}^{c]},
\end{aligned} \tag{273}$$

we have the desired result for the connection

$$\omega_\mu{}^{ac} = -2e^{\rho[a}\partial_{[\mu}e_{\rho]}^{c]} + e_{\mu b}e^{\rho a}e^{\nu c}\partial_{[\rho}e_{\nu]}^b, \tag{274}$$

and $b \leftrightarrow c$ to have the expression in the assignment

$$\omega_\mu{}^{ab} = -2e^{\rho[a}\partial_{[\mu}e_{\rho]}^{b]} + e_{\mu c}e^{\rho a}e^{\nu b}\partial_{[\rho}e_{\nu]}^c. \tag{275}$$

Appendix A. Black Hole parameters plot

The following Python script was used to generate the plots detailing the roots of the polynomials that define the radial locations of the event horizons.

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 plt.rcParams.update({
5     "text.usetex": False,
6     "mathtext.fontset": "cm",
7     "font.family": "serif",
8     "axes.titlesize": 14,
9     "axes.labelsize": 12,
10    "legend.fontsize": 10
11 })
12
13 # Parametros
14 M0 = 1.0
15 Lambda_DS = 0.3
16 Lambda_AdS = -0.6
17 M_crit = 1.0 / (3.0 * np.sqrt(Lambda_DS))
18
19 # Vectores r
20 r_flat = np.linspace(0, 4, 400)
21 r_ds = np.linspace(0, 4, 400)
22 r_ads = np.linspace(0, 3.5, 400)
23
24 # Figura y ejes
25 fig, axs = plt.subplots(1, 3, figsize=(15, 5))
26
27 # --- Panel 1: Lambda = 0 ---
28 axs[0].plot(r_flat, 1*r_flat - 2*M0, 'b-', label='k = +1')
29 axs[0].plot(r_flat, 0*r_flat - 2*M0, 'r-', label='k = 0')
30 axs[0].plot(r_flat, -1*r_flat - 2*M0, 'g-', label='k = -1')
31 axs[0].set_title(r'$\Lambda = 0$')
32 axs[0].set_ylabel(r'$P(r)$')
33 axs[0].set_ylim(-3, 3)
34
35 # --- Panel 2: Lambda > 0 ---
36 P_ds_sub = (Lambda_DS/3)*r_ds**3 - r_ds + 2*(0.4*M_crit)
37 P_ds_crit = (Lambda_DS/3)*r_ds**3 - r_ds + 2*(M_crit)
38 P_ds_super = (Lambda_DS/3)*r_ds**3 - r_ds + 2*(1.4*M_crit)
39
40 axs[1].plot(r_ds, P_ds_sub, 'b-', label=r'$M < M_{crit}$')
41 axs[1].plot(r_ds, P_ds_crit, 'r--', label=r'$M = M_{crit}$')
42 axs[1].plot(r_ds, P_ds_super, 'g-', label=r'$M > M_{crit}$')
43 axs[1].set_title(r'$\Lambda > 0, k = +1$')
44 axs[1].set_ylabel(r'$P(r)$')
45 axs[1].set_ylim(-1.5, 3)
46
47 # --- Panel 3: Lambda < 0 ---
48 axs[2].plot(r_ads, (abs(Lambda_AdS)/3)*r_ads**3 + 1*r_ads - 2*M0, 'b-', label='k = +1')
49 axs[2].plot(r_ads, (abs(Lambda_AdS)/3)*r_ads**3 + 0*r_ads - 2*M0, 'r-', label='k = 0')
50 axs[2].plot(r_ads, (abs(Lambda_AdS)/3)*r_ads**3 - 1*r_ads - 2*M0, 'g-', label='k = -1')
51 axs[2].set_title(r'$\Lambda < 0$')
52 axs[2].set_ylabel(r'$p(r)$')
53 axs[2].set_ylim(-3, 4)

```



```

54
55 # Configuración visual común para los tres paneles
56 for ax in axs:
57     ax.set_xlim(left=0)
58     ax.axhline(0, color='black', linewidth=0.8) # Eje x
59     ax.axvline(0, color='black', linewidth=0.8) # Eje y
60     ax.set_xlabel(r'$r$')
61     ax.legend(loc='lower center', bbox_to_anchor=(0.5, -0.25), ncol=3,
62             frameon=False)
63     # Ocultar bordes superior y derecho
64     ax.spines['top'].set_visible(False)
65     ax.spines['right'].set_visible(False)
66     ax.spines['bottom'].set_position('zero')
67
68 plt.tight_layout()
69 plt.subplots_adjust(bottom=0.2)
70 plt.show()

```

Listing 1: Python code to evaluate and plot the black hole horizon polynomials $P(r)$ and $p(r)$.

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